

```
/* ===== 文件: App.jsx ===== */
import { Routes, Route } from 'react-router-dom'
import TopBar from './components/TopBar.jsx'
import BottomNav from './components/BottomNav.jsx'
import Home from './pages/Home.jsx'
import Game from './pages/Game.jsx'
import Reports from './pages/Reports.jsx'
import Me from './pages/Me.jsx'
export default function App() {
  return (
    <div className="app-shell">
      <Routes>
        <Route path="/" element={<><TopBar title="会计小当家" /><Home /></> } />
        <Route path="/learn" element={<><TopBar title="经营我的公司" /><Game /></> } />
        <Route path="/game" element={<><TopBar title="经营我的公司" /><Game /></> } />
        <Route path="/reports" element={<><TopBar title="财务报表" /><Reports /></> } />
        <Route path="/me" element={<><TopBar title="我的" /><Me /></> } />
      </Routes>
      <BottomNav />
    </div>
  )
}
/* ===== 文件: components/BottomNav.jsx ===== */
import { NavLink } from 'react-router-dom'
const items = [
  { to: '/', emoji: '🏠', label: '首页', end: true },
  { to: '/game', emoji: '🎮', label: '经营' },
  { to: '/reports', emoji: '📊', label: '报表' },
  { to: '/me', emoji: '👤', label: '我的' },
]
export default function BottomNav() {
  return (
    <nav className="bottom-nav">
      {items.map((it) => (
        <NavLink
          key={it.to}
          to={it.to}
          end={it.end}
          className={({ isActive }) => 'nav-item' + (isActive ? ' active' : '')}
        >
          <span className="emoji">{it.emoji}</span>
          <span>{it.label}</span>
        </NavLink>
      ))}
    </nav>
  )
}
/* ===== 文件: components/EntryAnimation.jsx ===== */
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import { useState } from 'react'
export default function EntryAnimation({ entries }) {
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const [show, setShow] = useState(true)
const total = entries.reduce((s, e) => s + (Number(e.amount) || 0), 0)
return (
  <div className="card">
    <div className="flex between center" style={{ marginBottom: 12 }}>
      <strong>□ 会计分录演示</strong>
      <button className="btn ghost" style={{ width: 'auto', padding: '6px 14px', fontSize: 13 }} onClick={() => setShow(t
        □ 重播
      </button>
    </div>
    {show ? (
      <>
        {entries.map((e, i) => (
          <div
            key={i}
            className={`entry-row ${e.side === 'debit' ? 'delay-1' : 'delay-2'}}`
          >
            <div className={`entry-side ${e.side}`}>
              <div className="flex between center">
                <span className="label">{e.side === 'debit' ? '借 DEBIT' : '贷 CREDIT'}</span>
              </div>
              <div className="flex between center" style={{ marginTop: 4 }}>
                <span className="acc">{e.account}</span>
                <span className="amt">
                  {Number(e.amount) ? '□' + e.amount.toLocaleString() + '万' : '—'}
                </span>
              </div>
            </div>
          </div>
        ))}
        <div className="entry-amount-line mt12">
          {total ? `借贷合计: □${total.toLocaleString()}万` : '有借必有贷 · 借贷必相等'}
        </div>
      </>
    ) : (
      <div style={{ color: 'var(--text-soft)', fontSize: 13 }}>
        点击「播放动画」，看一笔业务如何变成规范的借贷分录 □
      </div>
    )}
  </div>
)
}
/* ===== 文件: components/ReportView.jsx ===== */

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import { useMemo } from 'react'
const TYPE_META = {
  balance: { title: '□ 资产负债表', sub: '资产 = 负债 + 所有者权益' },
  income: { title: '□ 利润表', sub: '收入 - 费用 = 利润' },
  cashflow: { title: '□ 现金流量表', sub: '经营 + 投资 + 筹资 = 净增加' },
}
export default function ReportView({ report }) {
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if (!report) return null
const meta = TYPE_META[report.type] || TYPE_META.balance
const { balanced, totals } = useMemo(() => {
  if (report.type === 'balance') {
    const asset = report.rows.filter((r) => r.side === 'asset').reduce((s, r) => s + r.value, 0)
    const liability = report.rows.filter((r) => r.side === 'liability').reduce((s, r) => s + r.value, 0)
    const equity = report.rows.filter((r) => r.side === 'equity').reduce((s, r) => s + r.value, 0)
    return { balanced: Math.abs(asset - (liability + equity)) < 0.05, totals: { asset, liability, equity, right: liability } }
  }
  if (report.type === 'income') {
    const net = report.rows.find((r) => r.item === '净利润')
    const revenue = report.rows.find((r) => r.item === '营业收入')
    const expense = report.rows.find((r) => r.item === '减: 所得税费用')
    const profit = net ? net.value : 0
    const rev = revenue ? revenue.value : 0
    return { balanced: true, totals: { revenue: rev, expense: 0, profit } }
  }
  const net = report.rows.reduce((s, r) => s + r.value, 0)
  return { balanced: true, totals: { net } }
}, [report])
return (
  <div className="card" style={{ background: '#FFDF8' }}>
    <div style={{ fontWeight: 800, fontSize: 16 }}>{meta.title}</div>
    <div style={{ color: 'var(--text-soft)', fontSize: 12, marginTop: 2 }}>{meta.sub}</div>
    {report.type === 'balance' && (
      <div style={{ marginTop: 12 }}>
        <ReportTable title="资产" rows={report.rows.filter((r) => r.side === 'asset')} />
        <ReportTable title="负债" rows={report.rows.filter((r) => r.side === 'liability')} />
        <ReportTable title="所有者权益" rows={report.rows.filter((r) => r.side === 'equity')} />
        <div className="flex between center" style={{ marginTop: 8, fontWeight: 800 }}>
          <span>资产合计</span><span className="v">{fmt(totals.asset)}</span>
        </div>
        <div className="flex between center" style={{ fontWeight: 800 }}>
          <span>负债+权益合计</span><span className="v">{fmt(totals.right)}</span>
        </div>
      </div>
    )}
    {report.type === 'income' && (
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    <div style={{ marginTop: 12 }}>
      <ReportTable title="" rows={report.rows} />
      <div className="flex between center" style={{ marginTop: 8, fontWeight: 800, color: 'var(--accent-deep)' }}>
        <span>净利润</span><span className="v">{fmt(totals.profit)}</span>
      </div>
    </div>
  )}
  {report.type === 'cashflow' && (
    <div style={{ marginTop: 12 }}>
      <ReportTable title="" rows={report.rows} />
      <div className="flex between center" style={{ marginTop: 8, fontWeight: 800, color: 'var(--accent-deep)' }}>
        <span>现金净增加额</span><span className="v">{fmt(totals.net)}</span>
      </div>
    </div>
  )}

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    </div>
  </div>
)}
<div
  className="mt12"
  style={{
    padding: '8px 12px',
    borderRadius: 10,
    fontSize: 13,
    fontWeight: 700,
    background: balanced ? '#E4F8F1' : '#FFEDED',
    color: balanced ? 'var(--accent-deep)' : '#FF6B6B',
  }}
>
  {balanced ? '☐ 报表平衡，编制正确!' : '☐ 报表不平衡，请检查项目'}
</div>
</div>
)
}
function ReportTable({ title, rows }) {
  if (!rows.length) return null
  return (
    <div style={{ marginBottom: 6 }}>
      {title && <div style={{ fontWeight: 700, fontSize: 13, color: 'var(--primary-deep)', margin: '6px 0' }}>{title}</div>}
      {rows.map((r, i) => (
        <div key={i} className="flex between center" style={{ padding: r.emphasize ? '8px 0' : '6px 0', borderBottom: '1px solid #ccc' }}>
          <span>{r.item}</span>
          <span style={{ fontWeight: 700 }}>{fmt(r.value)}</span>
        </div>
      ))}
    </div>
  )
}

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```
function fmt(n) {
  const v = Number(n) || 0
  return (v < 0 ? '-' : '') + '□' + Math.abs(v).toLocaleString() + '万'
}
/* ===== 文件: components/Toast.jsx ===== */
import { useEffect } from 'react'
export default function Toast({ message, onClose, duration = 1600 }) {
  useEffect(() => {
    if (!message) return
    const t = setTimeout(onClose, duration)
    return () => clearTimeout(t)
  }, [message, duration, onClose])
  if (!message) return null
  return <div className="toast">{message}</div>
}
/* ===== 文件: components/TopBar.jsx ===== */
import { useUser } from '../store/UserContext.jsx'
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import { TIERS } from '../data/tiers.js'
export default function TopBar({ title }) {
  const { state } = useUser()
  const tier = TIERS.find((t) => t.id === state.tier)
  return (
    <div className="topbar">
      <div className="title">{title}</div>
      <div className="spacer" />
      <div className="pill level" title="当前职称">□ {tier?.name}</div>
      <div className="pill coins">□ {state.coins}</div>
    </div>
  )
}
/* ===== 文件: data/badges.js ===== */
export const BADGES = [
  { id: 'firstBlood', name: '初出茅庐', emoji: '□', desc: '创立第一家公司和完成第一笔业务' },
  { id: 'firstEntry', name: '分录达人', emoji: '□', desc: '完成第一笔会计分录' },
  { id: 'boss', name: '公司掌舵人', emoji: '□', desc: '经营公司并盈利通关' },
  { id: 'cfo', name: '财务总监', emoji: '□', desc: '完整走完公司创立到结账全流程' },
  { id: 'reportPro', name: '报表专家', emoji: '□', desc: '生成资产负债表/利润表' },
  { id: 'allClear', name: '通关王者', emoji: '□', desc: '体验全部公司类型经营' },
]
/* ===== 文件: data/companies.js ===== */
export const COMPANIES = [
  {
    id: 'shop',
    name: '小生意（奶茶店）',
    emoji: '□',
```

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difficulty: 1,
blurb: '街角奶茶店, 科目少、流程简单, 最适合零基础入门。',
initCash: 20,
economics: {
  rent: 0.3,           // 月房租
  salary: 0.5,         // 月工资
  interestRate: 0.005, // 借款月利率
  dealSize: 2,         // 单笔进货/销售基准规模
  margin: 0.4,         // 毛利率
},
fixedAssets: [{ name: '制冰机/封口机', value: 3, monthlyDep: 0.1 }],
purchase: (amt) => ([
  { side: 'debit', account: '库存商品', amount: amt },
  { side: 'credit', account: '银行存款', amount: amt },
]),
costOfSale: (amt) => ([
  { side: 'debit', account: '主营业务成本', amount: amt },
  { side: 'credit', account: '库存商品', amount: amt },
]),
revenueAccount: '主营业务收入',
costAccount: '主营业务成本',
projects: [
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  { id: 'milktea', name: '经典奶茶', emoji: '☐', blurb: '走量快、毛利高、单价低, 最适合练手。', econ: { margin: 0.45, d
  { id: 'bake', name: '烘焙甜品', emoji: '☐', blurb: '客单价更高、损耗大, 对库存周转要求高。', econ: { margin: 0.5, dea
  { id: 'fruittea', name: '鲜果茶', emoji: '☐', blurb: '原材料鲜度要求高, 损耗与采购节奏更难把控。', econ: { margin: 0.
},
},
{
  id: 'tech',
  name: '科技公司',
  emoji: '☐',
  difficulty: 3,
  blurb: '做软件订阅, 有研发、服务器摊销、股权融资, 分录更丰富。',
  initCash: 50,
  economics: {
    rent: 0.8, salary: 1.5, interestRate: 0.006, dealSize: 3, margin: 0.6,
  },
  fixedAssets: [{ name: '服务器/办公设备', value: 10, monthlyDep: 0.4 }],
  purchase: (amt) => ([
    { side: 'debit', account: '研发支出-材料', amount: amt },
    { side: 'credit', account: '银行存款', amount: amt },
  ]),
  costOfSale: (amt) => ([
    { side: 'debit', account: '主营业务成本', amount: amt },
    { side: 'credit', account: '库存商品', amount: amt },

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    ]),
    revenueAccount: '主营业务收入-订阅',
    costAccount: '主营业务成本',
    projects: [
      { id: 'saas', name: 'SaaS 订阅', emoji: '🏠', blurb: '高复购、近乎零边际成本，毛利天花板最高。', econ: { margin: 0.7,
      { id: 'game', name: '游戏发行', emoji: '🎮', blurb: '爆款波动大、流水高但分成与买量成本高。', econ: { margin: 0.55, d
      { id: 'ai', name: 'AI 工具', emoji: '🤖', blurb: '前沿赛道，研发与算力（服务器）投入重。', econ: { margin: 0.6, dealS
    ],
    capitalizeRnd: (amt) => ([
      { side: 'debit', account: '无形资产-软件', amount: amt },
      { side: 'credit', account: '研发支出-材料', amount: amt },
    ]),
    amortize: (amt) => ([
      { side: 'debit', account: '管理费用-摊销', amount: amt },
      { side: 'credit', account: '累计摊销', amount: amt },
    ]),
  },
  {
    id: 'factory',
    name: '工厂',
    emoji: '🏭',
    difficulty: 4,
    blurb: '重资产制造，原材料→生产成本→库存商品，折旧与人工是大头。',
    initCash: 100,
    economics: {
      rent: 1.5, salary: 3, interestRate: 0.005, dealSize: 6, margin: 0.25,
    },
  },
}
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fixedAssets: [{ name: '厂房设备', value: 40, monthlyDep: 1.5 }],
purchase: (amt) => ([
  { side: 'debit', account: '原材料', amount: amt },
  { side: 'credit', account: '银行存款', amount: amt },
]),
produce: (amt) => ([
  { side: 'debit', account: '生产成本', amount: amt },
  { side: 'credit', account: '原材料', amount: amt },
]),
costOfSale: (amt) => ([
  { side: 'debit', account: '主营业务成本', amount: amt },
  { side: 'credit', account: '库存商品', amount: amt },
]),
revenueAccount: '主营业务收入-产成品',
costAccount: '主营业务成本',
projects: [
  { id: 'auto', name: '汽车配件', emoji: '🚗', blurb: '订单大、账期长，应收应付与存货管理是难点。', econ: { margin: 0.2
  { id: 'cloth', name: '服装制造', emoji: '👕', blurb: '季节性强、库存跌价风险高，对周转要求极高。', econ: { margin: 0.3

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    { id: 'food', name: '食品加工', emoji: '🍲', blurb: '原料涨跌频繁、保质期短，损耗与采购节奏难。', econ: { margin: 0.2,
  },
},
{
  id: 'trade',
  name: '贸易公司',
  emoji: '🏢',
  difficulty: 3,
  blurb: '大宗商品买卖，常赊销赊购，应收应付与汇兑风险突出。',
  initCash: 60,
  economics: {
    rent: 0.6, salary: 1.0, interestRate: 0.005, dealSize: 4, margin: 0.3,
  },
  fixedAssets: [{ name: '仓库/车辆', value: 15, monthlyDep: 0.5 }],
  purchase: (amt, onCredit) => onCredit ? ([
    { side: 'debit', account: '库存商品', amount: amt },
    { side: 'credit', account: '应付账款', amount: amt },
  ]) : ([
    { side: 'debit', account: '库存商品', amount: amt },
    { side: 'credit', account: '银行存款', amount: amt },
  ]),
  costOfSale: (amt) => ([
    { side: 'debit', account: '主营业务成本', amount: amt },
    { side: 'credit', account: '库存商品', amount: amt },
  ]),
  revenueAccount: '主营业务收入',
  costAccount: '主营业务成本',
  projects: [
    { id: 'elec', name: '电子数码', emoji: '📱', blurb: '单价高、更新快、贬值猛，库存管理是命门。', econ: { margin: 0.28,
    { id: 'energy', name: '能源化工', emoji: '🏭', blurb: '大宗波动大、常赊销赊购，汇兑与价格风险突出。', econ: { margin:
    { id: 'grain', name: '农副产品', emoji: '🌾', blurb: '季节与气候影响大，收购资金密集、账期长。', econ: { margin: 0.25,
  ],
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},
{
  id: 'listed',
  name: '上市公司',
  emoji: '🏢',
  difficulty: 5,
  blurb: '股份制、多轮融资、合并报表、严格披露，最完整的会计图景。',
  initCash: 200,
  economics: {
    rent: 4, salary: 8, interestRate: 0.004, dealSize: 12, margin: 0.3,
  },
  fixedAssets: [{ name: '总部大楼/设备', value: 80, monthlyDep: 3 }],
  purchase: (amt) => ([

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    { side: 'debit', account: '库存商品', amount: amt },
    { side: 'credit', account: '银行存款', amount: amt },
  ]),
  costOfSale: (amt) => ([
    { side: 'debit', account: '主营业务成本', amount: amt },
    { side: 'credit', account: '库存商品', amount: amt },
  ]),
  revenueAccount: '主营业务收入',
  costAccount: '主营业务成本',
  projects: [
    { id: 'finance', name: '金融控股', emoji: '🏦', blurb: '资本运作频繁、表外业务多，合并报表最复杂。', econ: { margin: 0.25 },
    { id: 'infra', name: '基建地产', emoji: '🏗️', blurb: '重资产、长周期、借款与预收房款勾稽关系密。', econ: { margin: 0.25 },
    { id: 'retail', name: '连锁零售', emoji: '🏬', blurb: '门店多、流水碎、存货与现金管理网点分散。', econ: { margin: 0.25 }
  ],
  issueStock: (amt) => ([
    { side: 'debit', account: '银行存款', amount: amt },
    { side: 'credit', account: '股本', amount: amt },
  ]),
},
]
export function getCompany(id) {
  return COMPANIES.find((c) => c.id === id) || COMPANIES[0]
}
/* ===== 文件: data/story.js ===== */
export const STORY = [
  {
    chapter: 1,
    title: '第一章 · 创立公司',
    tier: 'primary',
    steps: [
      {
        npc: '工商局小姐姐：欢迎创业！先选你要开什么类型的公司吧，不同类型账目可不一样哦～',
        options: [], // 公司类型由主页/开局选择，此处仅对话
        teach: '不同行业用不同会计科目：小生意用"库存商品"，工厂用"原材料/生产成本"，科技公司有"研发费用"。',
      },
      {
        npc: '导师：启动资金怎么来？全部自己出，还是借一部分？这决定了你的"实收资本"和"短期借款"科目，也影响后续利息与抵税。',
      }
    ]
  }
]

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options: [
  {
    label: '全部自有资金（稳健，无利息负担）',
    action: { type: 'fund', own: 'full', borrow: 0 },
    teach: '自有资金投入：借 银行存款 / 贷 实收资本。没有借款，后续没有利息支出，也无利息抵税。',
  },
  {
    label: '自有资金 + 银行借款 30%（杠杆经营）',
  }
]

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        action: { type: 'fund', own: 'part', borrow: 0.3 },
        recommended: true,
        teach: '借款入账：借 银行存款 / 贷 短期借款。注意：借款次月起每月要计提利息（财务费用），但利息可在税前扣除、抵',
    },
    {
        label: '自有资金 + 银行借款 70%（高杠杆，高风险）',
        action: { type: 'fund', own: 'low', borrow: 0.7 },
        teach: '高杠杆放大收益也放大风险：利息抵税更明显，但每月利息吃掉利润，现金紧张时易资金链断裂、破产。',
    },
],
},
{
    npc: '税务局：你打算登记成哪种纳税人？这决定增值税怎么交——小规模按 3% 征收率、进项不能抵扣；一般人按 13%、进项可抵',
    options: [
        {
            label: '小规模纳税人（征收率3%，进项不可抵扣）',
            action: { type: 'taxType', tax: 'small' },
            recommended: true,
            teach: '小规模：增值税=销售额×3%，采购进项不能抵扣，核算简单、税负低（适合小本经营）。',
        },
        {
            label: '一般纳税人（税率13%，进项可抵扣）',
            action: { type: 'taxType', tax: 'general' },
            teach: '一般纳税人：增值税=销项-进项，采购拿到专票可抵扣，适合进项多、规模大的企业；但核算更复杂。',
        },
    ],
},
{
    npc: '导师：营业执照到手！你的"实收资本"正式记在账上了。接下来要租店面、买设备。',
    options: [],
    teach: '实收资本是股东投入的本钱，属于所有者权益，和公司借钱（负债）性质完全不同。',
},
],
},
{
    chapter: 2,
    title: '第二章 · 筹备开业',
    tier: 'primary',
    steps: [
        {
            npc: '房东：店面谈妥啦！房租怎么付你定——押一付一最省现金压力；要是一次多付几个月，我给你租金折扣，长期更划算。押金',
            options: [

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```

    {
        label: '押一付一（现金流压力最小，无折扣）',
        action: { type: 'rent', months: 1, discount: 0 },

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        teach: '少付：只预付 1 个月房租，现金流出最小、资金压力小；但没有折扣，单位租金贵。',
    },
    {
        label: '押一付三（预付3个月，享5%租金折扣）',
        action: { type: 'rent', months: 3, discount: 0.05 },
        recommended: true,
        teach: '适度预付：一次付 3 个月房租，房东给 5% 折扣，单价更便宜；代价是当下多占一笔现金（挂"预付账款-房租"，按月
    },
    {
        label: '押一付六（预付6个月，享12%租金折扣）',
        action: { type: 'rent', months: 6, discount: 0.12 },
        teach: '多付：一次付 6 个月，折扣最高(12%)，长期最省租金；但当下现金流出大，资金链要绷得住，否则易断。',
    },
    ],
},
{
    npc: '设备商：制冰机、封口机、电脑，一套下来要花钱，但能用很多年。',
    options: [
        {
            label: '购置固定资产',
            action: { type: 'fixed', ratio: 0.15 },
            teach: '设备是固定资产，不能一次性算费用，要按月计提折旧分摊到各期（权责发生制）。',
        },
    ],
},
},
{
    npc: '导师：工厂要建生产线、科技公司要立项研发——不同公司，筹备阶段的科目差别就出来了。',
    options: [],
    teach: '工厂筹备进"在建工程/固定资产"；科技公司研发可费用化(研发费用)或资本化(无形资产)。',
},
],
},
{
    chapter: 3,
    title: '第三章 · 日常经营',
    tier: 'primary',
    steps: [
        {
            npc: (s) => {
                const cn = s?.co?.name || '公司'
                const pe = s?.projectEmoji || ''
                const pn = s?.projectName || ''
                return `供应商：进货啦！现金付还是赊账？${cn}${pe ? `· ${pe}${pn}` : ''}常赊购形成"应付账款"。`
            },
            options: [
                {
                    label: '现金进货上架',

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        action: { type: 'purchase', amt: 2 },
        teach: '进货：借 库存商品(或原材料) / 贷 银行存款。商品还在手里，是资产不是费用。',
    },
    {
        label: '赊账进货（形成应付账款）',
        action: { type: 'purchaseCredit', amt: 2 },
        teach: '赊购：借 库存商品 / 贷 应付账款。欠别人的钱是负债，到期要还。',
    },
],
},
{
    npc: (s) => {
        const cn = s?.co?.name || '公司'
        const pid = s?.projectId
        const cid = s?.co?.id
        const pe = s?.projectEmoji || ''
        const pn = s?.projectName || ''
        if (cid === 'shop') return `顾客: ${pid === 'milktea' ? '来一杯经典奶茶' : pid === 'bake' ? '来一份烘焙甜品' : '来'}
        if (cid === 'tech') return `客户: ${pid === 'game' ? '游戏充值到账' : pid === 'ai' ? 'AI 模型调用结算' : 'SaaS 续'}
        if (cid === 'factory') return `采购商: ${pid === 'auto' ? '汽车配件' : pid === 'cloth' ? '服装面料' : '食品加工料'}
        if (cid === 'trade') return `买家: ${pid === 'elec' ? '电子数码' : pid === 'energy' ? '能源化工' : '农副产品'}已签
        if (cid === 'listed') return `客户: ${pid === 'finance' ? '金融服务费' : pid === 'infra' ? '工程款' : '连锁零售门}
        return `顾客: ${pn ? `(${pe})${pn}` : ''}成交！扫码支付。`
    },
    options: [
        {
            label: '卖货收款',
            action: { type: 'sale', amt: 3, cost: 1.2 },
            teach: '收款：借 银行存款 / 贷 主营业务收入、应交税费。同时系统自动结转成本：借 主营业务成本 / 贷 库存商品。',
        },
    ],
},
{
    npc: 'HR: 该发工资了，这是"应付职工薪酬"。',
    options: [
        {
            label: '计提并发放工资',
            action: { type: 'salary', amt: 0.5 },
            teach: '工资：借 管理费用-工资 / 贷 应付职工薪酬。先计提后发放，体现权责发生制。',
        },
    ],
},
{
    npc: (s) => s.co.id === 'tech'
        ? '市场/研发部: 要不要加大投入？高投入能放大后续营收，但要先掏真金白银；低投入稳一点，增长慢。'
        : '市场部: 要不要投一笔营销/门店推广？高投入能放大后续营收，但要先掏真金白银；低投入稳一点，增长慢。',
    options: [
        {

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        label: '高投入（前置成本，后续营收+25%）',
        action: { type: 'invest', kind: 'market', level: 'high' },

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        teach: '高投入：当期确认一笔营销投入（科技公司计入「研发费用」科目，其他公司计入「销售费用」），减少利润与现金，
    },
    {
        label: '低投入（无前置成本，营收小幅提升）',
        action: { type: 'invest', kind: 'market', level: 'low' },
        recommended: true,
        teach: '低投入：不花前置成本，后续营收小幅提升，更稳健，适合现金流紧张时。',
    },
],
],
},
],
{
    chapter: 4,
    title: '第四章 · 月末结账',
    tier: 'middle',
    steps: [
        {
            npc: '导师：到月末了！系统自动帮你做两件大事，这是真实会计的硬规矩。',
            options: [],
            teach: '月末自动：① 计提折旧（借 管理费用-折旧 / 贷 累计折旧）② 借款计息（借 财务费用-利息 / 贷 应付利息）。',
        },
        {
            npc: '导师：看你有没有借款——如果有，本月起每月利息都要计提，哪怕钱还没付出去！',
            options: [
                {
                    label: '执行月末结账（折旧+计息+工资房租）',
                    action: { type: 'monthEnd' },
                    teach: '权责发生制：费用按受益期确认，不看出没出现金。所以没付的利息也要先记上。',
                },
            ],
        },
    ],
},
{
    npc: '税务：算算增值税（销项-进项）和企业所得税（利润×25%）。',
    options: [
        {
            label: '计提税金',
            action: { type: 'tax' },
            teach: '增值税是价外税，企业所得税 = 应纳税所得额 × 25%，都计入"应交税费"。',
        },
    ],
},
],
{

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npc: '导师: 把收入费用结转到"本年利润", 这一月的成果就出来了。',
options: [
  {
    label: '结转本年利润',
    action: { type: 'closeProfit' },
    teach: '结转: 收入、费用转入本年利润, 贷方余额盈利、借方余额亏损。利润表由此而生。',
  },
],

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],
},
],
},
{
  chapter: 5,
  title: '第五章 · 编制报表',
  tier: 'middle',
  steps: [
    {
      npc: '导师: 期末了, 出三张表。先出资产负债表——记住: 资产 = 负债 + 所有者权益, 必须平!',
      options: [
        {
          label: '生成资产负债表',
          action: { type: 'report', which: 'balance' },
          teach: '资产负债表是时点表。货币资金、存货、固定资产(净值)在左; 借款、应付、实收资本、本年利润在右。',
        },
      ],
    },
    {
      npc: '导师: 再看利润表: 收入 - 成本 - 费用 - 税 = 净利润。',
      options: [
        {
          label: '生成利润表',
          action: { type: 'report', which: 'income' },
          teach: '利润表是时期表, 从上到下: 营业收入→营业利润→净利润。',
        },
      ],
    },
    {
      npc: '导师: 三张表要能互相印证。利润表的净利润, 最后会流进资产负债表的"未分配利润"。',
      options: [
        {
          label: '查看报表勾稽',
          action: { type: 'reportCheck' },
          teach: '勾稽关系: 净利润→所有者权益; 现金净变动→货币资金。不平就是做错了。',
        },
      ],
    },
  ],
},

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    },
  ],
},
{
  chapter: 6,
  title: '第六章 · 扩张与决策',
  tier: 'senior',
  steps: [
    {
      npc: '投资人：生意不错，要扩店吗？再借一笔，还是用利润滚雪球？',
      options: [
        {

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      label: '再借款扩张（加杠杆）',
      action: { type: 'expand', borrow: 5 },
      teach: '再借款：借 银行存款 / 贷 短期借款。扩规模也扩了风险——利息更重了。',
    },
    {
      label: '用留存收益扩张（稳健）',
      action: { type: 'expand', own: 5 },
      teach: '用利润再投入不增加负债，财务更稳健，但扩张速度慢。',
    },
  ],
},
{
  npc: '分析师：用杜邦分析法看看你的 ROE = 净利率 × 资产周转率 × 权益乘数。',
  options: [
    {
      label: '做财务分析与诊断',
      action: { type: 'analyze' },
      teach: '高杠杆(权益乘数大)能拉高ROE，但现金为负就破产。盈利质量看经营现金流。',
    },
  ],
},
{
  npc: '导师：恭喜走到这里！你已完整经历了"创立→经营→结账→报表→决策"的全流程。',
  options: [
    {
      label: '完成开业挑战',
      action: { type: 'next' },
      teach: '真实会计就是这样：每一笔业务都是分录，每个月都要结账，每年都要出表。你已经是入门会计了！接下来进入持续经营',
    },
  ],
},
},
],
},
},

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{
  chapter: 7,
  title: '第七章 · 持续经营',
  tier: 'senior',
  loop: true,
  steps: [
    {
      npc: '导师：开业只是开始！真正开公司要一个月接一个月地经营。每过一个月，你会经历"进货→卖货→发工资→缴税→月末结账"',
      options: [
        {
          label: '□ 开始第 1 个经营月',
          action: { type: 'loopStart' },
          teach: '循环经营：每个月都要做一遍完整业务，月末系统自动结账。坚持越久，越能体会"权责发生制"和"现金流"的较劲。丰',
        },
      ],
    },
  ],
},
{
  {
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```

      npc: '供应商：新的一月，补点货吧？现金付还是赊账随你。',
      options: [
        {
          label: '现金进货上架',
          action: { type: 'purchase' },
          teach: '进货：借 库存商品 / 贷 银行存款。商品在手里是资产，没卖出去不算费用。',
          demoEntries: [{ side: 'debit', account: '库存商品', amount: 2 }, { side: 'credit', account: '银行存款', amount: 2 }],
        },
        {
          label: '赊账进货（形成应付账款）',
          action: { type: 'purchaseCredit' },
          teach: '赊购：借 库存商品 / 贷 应付账款。欠的钱是负债，记得下月有钱要还。',
          demoEntries: [{ side: 'debit', account: '库存商品', amount: 2 }, { side: 'credit', account: '应付账款', amount: 2 }],
        },
      ],
    },
  ],
  {
    npc: (s) => {
      const cn = s?.co?.name || '公司'
      const pid = s?.projectId
      const cid = s?.co?.id
      const pe = s?.projectEmoji || ''
      const pn = s?.projectName || ''
      let line
      if (cid === 'shop') line = `顾客：老样子，${pid === 'milktea' ? '经典奶茶' : pid === 'bake' ? '烘焙甜品' : '鲜果茶'}
      else if (cid === 'tech') line = `客户：${pid === 'game' ? '游戏充值到账' : pid === 'ai' ? 'AI 调用结算' : 'SaaS 续费'}
      else if (cid === 'factory') line = `采购商：${pid === 'auto' ? '汽车配件' : pid === 'cloth' ? '服装面料' : '食品加工'}
      else if (cid === 'trade') line = `买家：${pid === 'elec' ? '电子数码' : pid === 'energy' ? '能源化工' : '农副产品'}

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    else if (cid === 'listed') line = `客户: ${pid === 'finance' ? '金融服务费' : pid === 'infra' ? '工程款' : '连锁零售'}
    else line = `顾客: ${pn ? `($ {pe} $ {pn})` : ''}成交! 扫码支付。`
    return `${line} 本年行情会影响你的营业额——丰年门庭若市, 歉年门可罗雀。`
  },
  options: [
    {
      label: '卖货收款',
      action: { type: 'sale' },
      teach: '收款: 借 银行存款 / 贷 主营业务收入、应交税费; 系统自动结转成本: 借 主营业务成本 / 贷 库存商品。丰年营业额高, 歉年营业额低。',
      demoEntries: [
        { side: 'debit', account: '银行存款', amount: 3 },
        { side: 'credit', account: '主营业务收入', amount: 3 },
        { side: 'credit', account: '应交税费-销项', amount: 0.1 },
        { side: 'debit', account: '主营业务成本', amount: 1.2 },
        { side: 'credit', account: '库存商品', amount: 1.2 },
      ],
    },
  ],
},
{
  npc: 'HR: 又到发工资的时候了, 这是"应付职工薪酬"。你也可以选择拖欠——但那是一条邪路。',
  options: [

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    {
      label: '计提并发放工资（正道）',
      action: { type: 'salary' },
      teach: '工资: 借 管理费用-工资 / 贷 应付职工薪酬。先计提后发放, 权责发生制。准时发工资, 企业信誉才好。',
      demoEntries: [{ side: 'debit', account: '管理费用-工资', amount: 0.5 }, { side: 'credit', account: '应付职工薪酬' }],
    },
    {
      label: '❑ 拖欠工资不发（邪道）',
      action: { type: 'evilSalary' },
      teach: '邪道: 当月省下工资成本, 但欠薪挂账, 劳动稽查随时可能爆发, 且邪道玩得越多越危险。',
    },
  ],
},
{
  npc: '财务: 本月有收入, 要到税务局申报缴纳了。依法纳税是义务, 但国家给了很多"合法节税"的优惠——用足政策是正道; 动歪脑筋是邪道。',
  options: [
    {
      label: '依法纳税 + 用足合法优惠（正道·合理避税）',
      action: { type: 'tax', plans: ['smallBenefit', 'rndDeduction', 'vatSmall'] },
      recommended: true,
      teach: '合理避税≠偷税! 合法筹划: ①小型微利优惠（年所得≤300万所得税仅5%）; ②研发费用加计扣除100%（花1元研发费抵2元应纳税所得额）',
    },
  ],
}

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        label: '只做研发加计扣除（科技公司专属轻量筹划）',
        action: { type: 'tax', plans: ['rndDeduction'] },
        teach: '研发加计：把研发费用在税前扣除翻倍，少缴所得税，鼓励创新，完全合法。注意：仅科技类企业（软件/游戏/AI 研
    },
    {
        label: '☐ 税务调账改低应纳税额（邪道）',
        action: { type: 'evilTaxAdjust', ratio: 0.5 },
        teach: '邪道·调账：做分录把本期应纳税额做低，账面少缴。但调账次数越多，被税务局稽查的概率越高；一旦查出，要补回
    },
    {
        label: '☐ 直接拖欠税款不缴（邪道）',
        action: { type: 'evilTaxOwe' },
        teach: '邪道·拖欠：税款挂账不缴，从拖欠之日起按日加收万分之五滞纳金（按月约1.5%），越拖越贵，且被稽查概率随拖欠
    },
    {
        label: '☐ 虚开发票冲成本（邪道）',
        action: { type: 'evilFakeInvoice' },
        teach: '邪道：虚增成本少缴税，是刑事红线，随时可能爆雷。邪道越多，出事概率越高。',
    },
    ],
},
{
    npc: '导师：本月业务做完，系统帮你结账（折旧+计息+房租摊销），然后看你要不要继续。',
    options: [
        {
            label: '执行月末结账',
            action: { type: 'monthEnd' },

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```

        teach: '月末自动：计提折旧、借款计息、摊销预付房租。费用按受益期确认，不看出没出现金。每年末会检测邪道是否爆雷。
    },
    ],
},
{
    npc: '股东会：今年赚到钱啦！要不要从税后利润里给股东分红？分红要代扣 20% 股息红利个税。分红能让股东落袋为安，但会减
    options: [
        {
            label: '☐ 分红 30%（落袋为安）',
            action: { type: 'dividend', ratio: 0.3 },
            teach: '分红：借 利润分配-应付股利、应付股利 / 贷 银行存款，同时代扣个税（借 应交税费-个人所得税）。分红后"累计
        },
        {
            label: '☐ 分红 50%（多分些）',
            action: { type: 'dividend', ratio: 0.5 },
            teach: '多分红：股东拿得更多，但公司留存现金更少，扩张与抗风险能力变弱。需在"回报股东"和"公司发展"间平衡。',
        },
    ],
}

```

```

        label: '☐ 暂不分红（利润留存再投入）',
        action: { type: 'dividend', ratio: 0 },
        teach: '不分红：利润留在公司（未分配利润），可用于下月扩张或应对歉年，财务更稳健，但股东当期没有现金回报。',
    },
],
},
{
  npc: '导师：这个月经营结束啦。是咬牙继续开下去，还是见好就收、结业清算？撑得越久规模越大，但现金一旦为负就破产。本年
  options: [
    {
      label: '☐ 继续经营下一个月',
      action: { type: 'loopContinue' },
      teach: '继续经营：时间往前走一个月，重复"进-销-薪-税-结"。撑得越久，累计利润和规模越可观，但现金一旦为负就破产。',
    },
    {
      label: '☐ 结业清算（出最终报表）',
      action: { type: 'close' },
      teach: '结业：把剩余资产变现、还清负债，算出最终净利润与现金，给出经营成绩单。',
    },
  ],
},
],
},
],
}
]
/* ===== 文件: data/tiers.js ===== */
export const TIERS = [
  { id: 'primary', name: '初级会计', color: 'var(--accent)', desc: '创立公司 · 凭证记账 · 资产负债表' },
  { id: 'middle', name: '中级会计', color: 'var(--primary)', desc: '成本结转 · 折旧计息 · 利润表' },
  { id: 'senior', name: '高级会计', color: 'var(--gold)', desc: '合并报表 · 财务分析 · 扩张决策' },
]
/* ===== 文件: lib/engine.js ===== */
import { getCompany } from '../data/companies.js'
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```

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```

export const fmtW = (v) => Number((v || 0).toFixed(1)).toFixed(1)
const INITIAL_BALANCES = () => ({
  银行存款: 0, 库存现金: 0, 应收账款: 0, 应付账款: 0, 原材料: 0, 库存商品: 0,
  生产成本: 0, 固定资产: 0, 累计折旧: 0, 累计摊销: 0, 无形资产: 0, 研发支出: 0,
  预付账款: 0, 短期借款: 0, 应付职工薪酬: 0, 应付利息: 0, 应交税费: 0,
  实收资本: 0, 股本: 0, 本年利润: 0, 主营业务收入: 0, 主营业务成本: 0,
  管理费用: 0, 财务费用: 0, 研发费用: 0, 所得税费用: 0,
})
const NATURE = {
  银行存款: true, 库存现金: true, 应收账款: true, 原材料: true, 库存商品: true,
  生产成本: true, 固定资产: true, 无形资产: true, 研发支出: true, 预付账款: true,
  主营业务成本: true, 管理费用: true, 财务费用: true, 研发费用: true, 所得税费用: true,
  短期借款: false, 应付账款: false, 应付职工薪酬: false, 应付利息: false, 应交税费: false,

```

```

    实收资本: false, 股本: false, 本年利润: false, 累计折旧: false, 累计摊销: false, 主营业务收入: false,
  }
function natureOf(account) {
  if (NATURE[account] !== undefined) return NATURE[account]
  const parent = account.split('-')[0]
  return NATURE[parent] !== undefined ? NATURE[parent] : true
}
function bump(balances, account, side, amount) {
  const isDebitNormal = natureOf(account)
  let delta = amount
  if (side === 'credit') delta = -delta
  if (!isDebitNormal) delta = -delta
  balances[account] = (balances[account] || 0) + delta
  return delta
}
function pnlCat(account) {
  if (account === '主营业务收入' || account.startsWith('主营业务收入')) return 'revenue'
  if (account === '主营业务成本' || account.startsWith('主营业务成本')) return 'cost'
  if (account === '管理费用' || account.startsWith('管理费用')) return 'mgmt'
  if (account === '财务费用' || account.startsWith('财务费用')) return 'fin'
  if (account === '研发费用' || account.startsWith('研发费用')) return 'rd'
  if (account === '所得税费用' || account.startsWith('所得税费用')) return 'tax'
  return null
}
export const DIFFICULTY = {
  easy: { id: 'easy', name: '简单模式', lives: 3, autoFix: true, randFund: false, selfEntry: false,
    hint: '系统给出分录, 错了自动修正, 容错 3 次' },
  hard: { id: 'hard', name: '困难模式', lives: 1, autoFix: false, randFund: true, selfEntry: true,
    hint: '自己写分录, 金额随机, 错 1 次扣容错, 必须做调整分录' },
  hardcore: { id: 'hardcore', name: '硬核模式', lives: 0, autoFix: false, randFund: true, selfEntry: true,
    hint: '错 1 次即失败, 学不到最终关' },
}
export const TAX = {
  VAT_SMALL: 0.03, // 小规模纳税人征收率
  VAT_GENERAL: 0.13, // 一般纳税人税率（货物）
  CIT: 0.25, // 企业所得税标准税率
  CIT_SMALL: 0.05, // 小型微利企业优惠（年应税所得额≤300万部分, 实际税负5%）
}
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```

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```

  RND_SUPER: 1.0, // 研发费用加计扣除比例（100%, 即每花1元研发, 税前扣除2元）
  FORCE_THRESHOLD: 500, // 年应税销售额超 500万 强制登记为一般纳税人
}
export const TAX_PLANS = {
  smallBenefit: { id: 'smallBenefit', name: '小型微利优惠', desc: '年应税所得≤300万, 所得税按5%而非25%' },
  rndDeduction: { id: 'rndDeduction', name: '研发加计抵扣', desc: '研发费用加计扣除100%, 税前扣除翻倍' },
  vatSmall: { id: 'vatSmall', name: '小规模免税', desc: '小规模纳税人季度销售额≤30万免征增值税' },
}

```

```

export function createCompany(companyId, diffId = 'easy', projectId = null) {
  const base = getCompany(companyId)
  const projects = base.projects || []
  const proj = projects.find((p) => p.id === projectId) || projects[0] || null
  const co = proj && proj.econ
    ? { ...base, economics: { ...base.economics, ...proj.econ } }
    : base
  const balances = INITIAL_BALANCES()
  const state = {
    companyId, co, month: 0, balances, ledger: [], loans: [],
    vouchers: [], errors: [], lives: DIFFICULTY[diffId].lives, difficulty: diffId,
    penalty: 0, history: [], failed: false, failedReason: '', scale: 1,
    taxType: 'small', cumSales: 0, vatOutput: 0, vatInput: 0,
    forcedGeneral: false, boost: 0, choices: {},
    projectId: proj ? proj.id : null,
    projectName: proj ? proj.name : null,
    projectEmoji: proj ? proj.emoji : null,
    year: 1, // 当前经营年份（1 起）
    yearMood: 'normal', // 本年行情: good 丰年 / bad 歉年 / normal
    evilCount: 0, // 累计"邪道玩法"次数
    evilEvents: [], // 已埋下的雷: {atYear, type}（恶果延迟爆发）
    skippedTaxMonths: 0, // 累计未缴税月数（用于爆雷时补缴+罚款）
    wageDiscontent: 0, // 隐藏不满度 0~100（拖欠工资累积，影响效率）
    wageStrikes: 0, // 罢工黑历史次数（越多越易再罢工、损失越大）
    wageUnpaid: 0, // 累计欠薪额（万）
    strikeThisMonth: false, // 本月是否发生罢工（影响本月效率与成本）
    pendingStrike: null, // 罢工待处理: {month} → UI 弹窗三选一
    taxAdjusts: 0, // 调账改应纳税额次数（产生分录，按次数定稽查概率）
    taxAdjustAmount: 0, // 累计调账虚减的税额（被查到需补回）
    taxOwed: 0, // 直接拖欠的累计税额（产生每日万分之五滞纳金）
    taxLateFee: 0, // 累计滞纳金（万）
    auditHits: 0, // 被税务局查处次数（黑历史）
    totalDividend: 0, // 累计已分红净额（股东实得）
    quarterRevenue: 0, // 本季度累计营收（用于小规模免税判定）
    usedTaxPlans: [], // 已采用的合法筹划手段 id 列表
    cum: { revenue: 0, cost: 0, mgmt: 0, fin: 0, rd: 0, tax: 0 }, // 累计损益（实时追踪，供利润表展示）
    receivables: [], // 应收账款（赊销，F）
    payablesDue: [], // 到期应付账款（F）
    reachedMilestones: [], // 已达成里程碑 id（D）
    choiceLog: [], // 决策记录（B）
    eventBuff: {}, // 当月随机事件数值缓冲（C）
    monthOrders: [], // 当月订单（E）
  }
}
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```

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```

    lastEvent: null, // 最近一次随机事件（C）
  }
  return state

```

```

}
export function applyFunding(state, choice) {
  const co = state.co
  const own = co.initCash // 本金（股东出资）固定
  let borrow
  if (choice === 'low') borrow = +(own * 0.7).toFixed(1)
  else if (choice === 'part') borrow = +(own * 0.3).toFixed(1)
  else borrow = 0
  state.balances['银行存款'] = +(own + borrow).toFixed(1) // 可动用资金 = 本金 + 借款
  state.balances['实收资本'] = own
  state.balances['短期借款'] = borrow
  state.scale = +((own + borrow) / own).toFixed(3) // 借款放大经营规模
  state.loans = borrow > 0 ? [{ principal: borrow, rate: co.economics.interestRate, since: 1 }] : []
  state.choices.leverage = choice
  return state
}
export function applyExpand(state, { borrow = 0, own = 0 } = {}) {
  const co = state.co
  if (borrow > 0) {
    applyBusiness(state, mk('银行存款', borrow, '短期借款', borrow, '扩张：银行借款'), state.month)
    state.loans.push({ principal: borrow, rate: co.economics.interestRate, since: state.month + 1 })
  }
  if (own > 0) {
    applyBusiness(state, mk('银行存款', own, '实收资本', own, '扩张：利润再投入'), state.month)
  }
  state.scale = +((state.scale || 1) + (borrow + own) / co.initCash).toFixed(3)
  return state
}
export function applyTaxType(state, type) {
  state.taxType = type === 'general' ? 'general' : 'small'
  state.choices.taxType = state.taxType
  return state
}
export function vatOnSale(state, saleAmt, revenueAccount) {
  const rate = state.taxType === 'general' ? TAX.VAT_GENERAL : TAX.VAT_SMALL
  const vat = +(saleAmt * rate).toFixed(2)
  state.cumSales = +(state.cumSales + saleAmt).toFixed(2)
  state.vatOutput = +(state.vatOutput + vat).toFixed(2)
  const entries = [
    { side: 'debit', account: '银行存款', amount: +(saleAmt + vat).toFixed(2) },
    { side: 'credit', account: revenueAccount, amount: saleAmt },
    { side: 'credit', account: '应交税费-销项', amount: vat },
  ]
  return { entries, vat, saleAmt, rate }
}
export function vatOnPurchase(state, purAmt, onCredit) {
  const rate = state.taxType === 'general' ? TAX.VAT_GENERAL : 0

```

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```
const vat = +(purAmt * rate).toFixed(2)
state.vatInput = +(state.vatInput + vat).toFixed(2)
const debit = [{ side: 'debit', account: '库存商品', amount: purAmt }]
if (vat > 0) debit.push({ side: 'debit', account: '应交税费-进项', amount: vat })
const credit = onCredit
  ? [{ side: 'credit', account: '应付账款', amount: +(purAmt + vat).toFixed(2) }]
  : [{ side: 'credit', account: '银行存款', amount: +(purAmt + vat).toFixed(2) }]
return { entries: [...debit, ...credit], vat }
}

export function maybeForceGeneral(state) {
  if (state.taxType === 'small' && state.cumSales > TAX.FORCE_THRESHOLD) {
    state.taxType = 'general'
    state.forcedGeneral = true
    state.choices.taxType = 'general'
    return true
  }
  return false
}

export function applyInvest(state, kind, level) {
  state.choices[kind] = level
  if (kind === 'market') {
    if (level === 'high') {
      const cost = +(state.co.economics.dealSize * (state.scale || 1) * 0.5).toFixed(1)
      applyBusiness(state, mk('银行存款', cost, '管理费用-营销', cost, '高投入营销'), state.month)
      state.boost = (state.boost || 0) + 0.25
    } else {
      state.boost = (state.boost || 0) + 0.05
    }
  }
  return state
}

export function settleTax(state, plans = []) {
  const planSet = new Set(plans)
  let vatPayable = state.taxType === 'general'
    ? +(state.vatOutput - state.vatInput).toFixed(2)
    : +state.vatOutput.toFixed(2)
  vatPayable = Math.max(0, vatPayable)
  if (state.taxType !== 'general' && planSet.has('vatSmall') && vatPayable > 0 && (state.quarterRevenue || 0) <= 30) {
    vatPayable = 0
    plans = plans.filter((p) => p !== 'vatSmall')
    planSet.delete('vatSmall')
  }
  if (planSet.has('rndDeduction') && state.co?.id !== 'tech') {
    planSet.delete('rndDeduction')
    plans = plans.filter((p) => p !== 'rndDeduction')
  }
  let taxableProfit = state.balances['本年利润'] || 0
  if (planSet.has('rndDeduction')) {
```

```

    const rnd = state.balances['研发费用'] || 0
    taxableProfit = +(taxableProfit - rnd).toFixed(2) // 加计100%即再扣一次研发费用
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```

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```

}
const useSmallBenefit = planSet.has('smallBenefit') && taxableProfit > 0 && taxableProfit <= 300
const citRate = useSmallBenefit || (taxableProfit > 0 && taxableProfit <= 300) ? TAX.CIT_SMALL : TAX.CIT
const cit = +(Math.max(0, taxableProfit) * citRate).toFixed(2)
const entries = []
if (vatPayable > 0) {
  entries.push(...mk('应交税费-销项', state.vatOutput, '应交税费-未交增值税', state.vatOutput, '结转销项税额'))
  if (state.taxType === 'general' && state.vatInput > 0) {
    entries.push(...mk('应交税费-未交增值税', state.vatInput, '应交税费-进项', state.vatInput, '结转进项税额'))
  }
  entries.push(...mk('应交税费-未交增值税', vatPayable, '银行存款', vatPayable, '缴纳增值税'))
}
if (cit > 0) {
  entries.push(...mk('所得税费用', cit, '应交税费-所得税', cit, '计提企业所得税' + (useSmallBenefit ? '（小微优惠）' : '')))
  entries.push(...mk('本年利润', cit, '所得税费用', cit, '结转所得税费用至本年利润'))
  entries.push(...mk('应交税费-所得税', cit, '银行存款', cit, '缴纳企业所得税'))
}
applyBusiness(state, entries, '缴纳税金' + (plans.length ? '（含合法筹划）' : ''), state.month)
state.vatOutput = 0; state.vatInput = 0
state.quarterRevenue = 0
return { vatPayable, cit, taxType: state.taxType, forced: state.forcedGeneral, plans: plans.filter((p) => planSet.has(p)) }
}
export function declareDividend(state, ratio = 0.3) {
  const profit = state.balances['本年利润'] || 0
  if (profit <= 0) {
    return { ok: false, amount: 0, msg: '本年无可供分配的利润，无法分红' }
  }
  const div = +(profit * ratio).toFixed(2)
  if (div <= 0) return { ok: false, amount: 0, msg: '分红金额为0' }
  const divTax = +(div * 0.2).toFixed(2)
  const net = +(div - divTax).toFixed(2)
  const entries = [
    ...mk('应付股利', div, '银行存款', div, `宣告并支付股东分红${Math.round(ratio * 100)}%`),
    ...mk('利润分配-应付股利', div, '应付股利', div, '结转应付股利'),
    ...mk('应交税费-个人所得税', divTax, '银行存款', divTax, '代扣股息红利个税20%'),
  ]
  applyBusiness(state, entries, `股东分红（${Math.round(ratio * 100)}%）`, state.month)
  state.totalDividend = +(state.totalDividend || 0) + net
  return { ok: true, amount: net, gross: div, tax: divTax, msg: `分红 ${fmtW(net)}万（代扣个税 ${fmtW(divTax)}万）` }
}
function mk(d, da, c, ca, desc) {
  return [
    { side: 'debit', account: d, amount: da, desc },

```

```

    { side: 'credit', account: c, amount: ca, desc },
  ]
}
function withMeta(e, month, i) {
  return { desc: e.desc || '', debit: e.side === 'debit' ? e.account : '', credit: e.side === 'credit' ? e.account : '', am
}
export function applyBusiness(state, entries, desc, month) {
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```

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```

const ledger = [...state.ledger]
entries.forEach((e) => {
  const delta = bump(state.balances, e.account, e.side, e.amount)
  const cat = pnlCat(e.account)
  if (cat && state.cum) state.cum[cat] = +((state.cum[cat] || 0) + delta).toFixed(2)
})
entries.forEach((e, i) => ledger.push(withMeta(e, month ?? state.month, ledger.length + i)))
state.ledger = ledger
return state
}
export function recordVoucher(state, { desc, actual, expected, month }) {
  const correct = gradeEntries(actual, expected)
  const v = { id: state.vouchers.length + 1, month: month ?? state.month, desc, actual, expected, correct }
  state.vouchers.push(v)
  if (!correct) state.errors.push(v.id)
  return v
}
export function gradeEntries(actual, expected) {
  if (!actual || actual.length !== expected.length) return false
  const norm = (list) => list.map((e) => ({ side: e.side, account: e.account, amount: +e.amount })))
    .sort((a, b) => (a.side + a.account).localeCompare(b.side + b.account) || a.amount - b.amount)
  const a = norm(actual), b = norm(expected)
  return a.every((e, i) => e.side === b[i].side && e.account === b[i].account && Math.abs(e.amount - b[i].amount) < 0.001)
}
export function monthEnd(state, opts = {}) {
  const co = state.co
  const m = state.month + 1
  state.month = m
  const newEntries = []
  co.fixedAssets.forEach((fa) => newEntries.push(...mk('管理费用-折旧', fa.monthlyDep, '累计折旧', fa.monthlyDep, `计提${fa.
state.loans.forEach((loan) => {
  const interest = +(loan.principal * loan.rate).toFixed(2)
  if (m >= loan.since) newEntries.push(...mk('财务费用-利息', interest, '应付利息', interest, `计提借款利息(月${m})`))
})
const prepRent = state.choices?.rentMonths
if (prepRent && prepRent > 0 && (state.balances['预付账款-房租'] || 0) > 0) {
  const grossMonthly = co.economics.rent * (state.scale || 1)
  const discount = state.choices.rentDiscount || 0

```

```

    const monthlyRent = +(grossMonthly * (1 - discount)).toFixed(2)
    const remain = state.balances['预付账款-房租']
    const amort = remain >= monthlyRent ? monthlyRent : remain
    if (amort > 0) {
        newEntries.push(...mk('管理费用-房租', amort, '预付账款-房租', amort, `摊销预付房租(月${m})`))
        state.balances['预付账款-房租'] = +(remain - amort).toFixed(2)
    }
}
}
applyBusiness(state, newEntries, `第${m}月末结账`, m)
const rev = sumAccount(state.balances, '主营业务收入')
const costs = sumAccount(state.balances, '主营业务成本') + sumAccount(state.balances, '管理费用') +
    sumAccount(state.balances, '财务费用') + sumAccount(state.balances, '研发费用') + sumAccount(state.balances, '所得税费用')
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```

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```

const monthProfit = +(rev - costs).toFixed(2)
state.balances['本年利润'] = +((state.balances['本年利润'] || 0) + monthProfit).toFixed(2)
for (const k of ['主营业务收入', '主营业务成本', '管理费用', '财务费用', '研发费用', '所得税费用', '管理费用-折旧', '管理费用-摊销']) {
    if (state.balances[k] != null) state.balances[k] = 0
}
state.history.push({ month: m, cash: state.balances['银行存款'], profit: state.balances['本年利润'], revenue: rev, totalAssets: state.balances['总资产'] })
if (m > 1 && m % 12 === 1) {
    state.year = (state.year || 1) + 1
    state.yearMood = rollYearMood()
}
accrueTaxLateFee(state)
rollStrike(state)
taxAudit(state, m)
maybeEvilBlowup(state, m)
return state
}
export function settleErrors(state) {
    const diff = DIFFICULTY[state.difficulty]
    const bad = state.vouchers.filter((v) => !v.correct && v.adjustDone !== true && !v.pardoned)
    if (bad.length === 0) return { tasks: [], penalized: 0 }
    const penalized = bad.length * (state.difficulty === 'hardcore' ? 0 : 0.5) // 罚款（硬核直接失败）
    let tasks = []
    if (diff.autoFix) {
        bad.forEach((v) => {
            v.pardoned = true
            applyBusiness(state, v.expected, `自动修正: ${v.desc}`, state.month)
        })
    }
    if (state.difficulty === 'hardcore') {
        state.failed = true
        state.failedReason = '硬核模式：凭证记错一次即失败'
        return { tasks: [], penalized: 0, fatal: true }
    }
    state.balances['银行存款'] = +(state.balances['银行存款'] - penalized).toFixed(2)
}

```

```

    state.penalty += penalized
    tasks = bad.map((v) => ({
      vid: v.id,
      desc: v.desc,
      wrong: v.actual,
      hint: `原业务应记为: ${v.expected.map((e) => `${e.side === 'debit' ? '借' : '贷'} ${e.account} ${e.amount}`).join(' / ')}
      done: false,
    })))
  }
  return { tasks, penalized }
}

export function applyAdjust(state, vid, entries) {
  const v = state.vouchers.find((x) => x.id === vid)
  if (!v) return false
  applyBusiness(state, entries, `调整分录: 修正${v.desc}`, state.month)
  v.adjustDone = true
  return true
}
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```

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```

}

export function loseLife(state) {
  state.lives -= 1
  if (state.lives < 0) {
    state.failed = true
    state.failedReason = '容错次数用尽, 经营失败'
  }
  return state.lives
}

export function totalAssets(b) {
  return +(b['银行存款'] + b['库存现金'] + b['应收账款'] + b['原材料'] + b['库存商品'] +
    b['生产成本'] + b['固定资产'] - b['累计折旧'] + b['无形资产'] - b['累计摊销'] + b['研发支出'] + b['预付账款']).toFixed(2)
}

export function sumAccount(b, parent) {
  let s = 0
  for (const k of Object.keys(b)) if (k === parent || k.startsWith(parent + '-')) s += (b[k] || 0)
  return +s.toFixed(2)
}

export function isBankrupt(state) {
  const assets = totalAssets(state.balances)
  const debt = sumAccount(state.balances, '短期借款') + sumAccount(state.balances, '应付账款') +
    sumAccount(state.balances, '应付职工薪酬') + sumAccount(state.balances, '应付利息') + sumAccount(state.balances, '应交税费')
  return assets < debt || state.balances['银行存款'] < 0
}

export function buildReports(state) {
  const b = state.balances || {}
  const g = (k) => sumAccount(b, k)
  const assets = [

```

```

    { item: '货币资金', value: +(g('银行存款') + g('库存现金')).toFixed(2), side: 'asset' },
    { item: '应收账款', value: +g('应收账款').toFixed(2), side: 'asset' },
    { item: '存货(含原材料/库存/在产)', value: +(g('原材料') + g('库存商品') + g('生产成本')).toFixed(2), side: 'asset' },
    { item: '预付账款', value: +g('预付账款').toFixed(2), side: 'asset' },
    { item: '固定资产(净值)', value: +(g('固定资产') - g('累计折旧')).toFixed(2), side: 'asset' },
    { item: '无形资产(净值)', value: +(g('无形资产') - g('累计摊销')).toFixed(2), side: 'asset' },
    { item: '研发支出', value: +g('研发支出').toFixed(2), side: 'asset' },
  ]
  const liabilities = [
    { item: '短期借款', value: +g('短期借款').toFixed(2), side: 'liability' },
    { item: '应付账款', value: +g('应付账款').toFixed(2), side: 'liability' },
    { item: '应付职工薪酬', value: +g('应付职工薪酬').toFixed(2), side: 'liability' },
    { item: '应付利息', value: +g('应付利息').toFixed(2), side: 'liability' },
    { item: '应交税费', value: +g('应交税费').toFixed(2), side: 'liability' },
  ]
  const equity = [
    { item: '实收资本/股本', value: +(g('实收资本') + g('股本')).toFixed(2), side: 'equity' },
    { item: '本年利润', value: +g('本年利润').toFixed(2), side: 'equity' },
  ]
  const balance = { type: 'balance', rows: [...assets, ...liabilities, ...equity] }
  const cum = state.cum || {}
  const revenue = +(cum.revenue !== null ? cum.revenue : sumAccount(b, '主营业务收入')).toFixed(2)

```

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```

const cost = +(cum.cost !== null ? cum.cost : sumAccount(b, '主营业务成本')).toFixed(2)
const mgmt = +(cum.mgmt !== null ? cum.mgmt : sumAccount(b, '管理费用')).toFixed(2)
const fin = +(cum.fin !== null ? cum.fin : sumAccount(b, '财务费用')).toFixed(2)
const rd = +(cum.rd !== null ? cum.rd : sumAccount(b, '研发费用')).toFixed(2)
const tax = +(cum.tax !== null ? cum.tax : sumAccount(b, '所得税费用')).toFixed(2)
const netProfit = +(state.balances['本年利润'] || 0).toFixed(2) // 累计净利润以「本年利润」为准（带符号）
const income = {
  type: 'income',
  rows: [
    { item: '营业收入', value: revenue },
    { item: '减：营业成本', value: -cost },
    { item: '减：管理费用', value: -mgmt },
    { item: '减：财务费用', value: -fin },
    { item: '减：研发费用', value: -rd },
    { item: '减：所得税费用', value: -tax },
    { item: '净利润', value: netProfit, emphasize: true },
  ],
}
const np = +g('本年利润').toFixed(2)
const dep = +g('累计折旧').toFixed(2)
const amort = +g('累计摊销').toFixed(2)
const monthPnL = +(g('主营业务收入') - g('主营业务成本') - g('管理费用') - g('财务费用') - g('研发费用') - g('所得税费用')).toFixed(2)
const inventory = +(g('库存商品') + g('原材料') + g('生产成本')).toFixed(2) // 存货占用（期初0）

```

```

const receivable = +g('应收账款').toFixed(2) // 应收占用（期初0）
const payable = +(g('应付账款') + g('应付职工薪酬') + g('应交税费') + g('应付利息')).toFixed(2) // 应付增加（期初0）
const opNet = +(np + monthPnL + dep + amort - inventory - receivable + payable).toFixed(2)
const invOut = -(+g('固定资产').toFixed(2) + +g('无形资产').toFixed(2) + +g('研发支出').toFixed(2)) // 购建长期资产（期初0）
const invNet = +invOut.toFixed(2)
const finIn = +(g('短期借款') + g('实收资本') + g('股本')).toFixed(2) // 借款+出资流入（期初0）
const finOut = -(+g('应付股利').toFixed(2) + (state.dividendPaid || 0)) // 分红流出
const finNet = +(finIn + finOut).toFixed(2)
const cashNet = +(opNet + invNet + finNet).toFixed(2)
const endCash = +(g('银行存款') + g('库存现金')).toFixed(2)
const cashflow = {
  type: 'cashflow',
  note: '间接法（基于累计余额，与资产负债表勾稽）',
  rows: [
    { item: '一、经营活动现金流量', level: 'h' },
    { item: '净利润', value: np },
    { item: '加：累计折旧', value: dep },
    { item: '加：累计摊销', value: amort },
    { item: '减：存货的增加', value: -inventory },
    { item: '减：应收账款的增加', value: -receivable },
    { item: '加：应付账款等经营性负债的增加', value: payable },
    { item: '经营活动产生的现金流量净额', value: opNet, level: 'total' },
    { item: '二、投资活动现金流量', level: 'h' },
    { item: '购建固定资产、无形资产等支付的现金', value: invOut },
    { item: '投资活动产生的现金流量净额', value: invNet, level: 'total' },
    { item: '三、筹资活动现金流量', level: 'h' },
    { item: '取得借款、吸收投资收到的现金', value: finIn },

```

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```

    { item: '偿还债务、分配利润支付的现金', value: finOut },
    { item: '筹资活动产生的现金流量净额', value: finNet, level: 'total' },
    { item: '四、现金及现金等价物净增加额', value: cashNet, level: 'total' },
    { item: '期末现金及现金等价物余额', value: endCash, level: 'total' },
  ],
}
return { balance, income, cashflow }
}
export function rollYearMood() {
  const r = Math.random()
  if (r < 0.33) return 'good'
  if (r < 0.66) return 'bad'
  return 'normal'
}
export const MOOD_LABEL = { good: '☐ 丰年', bad: '☐ 歉年', normal: '☐ 平常年' }
export function moodFactor(state) {
  return state.yearMood === 'good' ? 1.25 : state.yearMood === 'bad' ? 0.7 : 1.0
}

```

```

export function investEff(state) {
  return state.yearMood === 'good' ? 1.4 : state.yearMood === 'bad' ? 0.5 : 1.0
}
export function evilAct(state, kind) {
  if (kind === 'salary') return evilSalary(state)
  if (kind === 'taxAdjust') return evilTaxAdjust(state)
  if (kind === 'taxOwe') return evilTaxOwe(state)
  if (kind === 'fakeInvoice') {
    state.evilmCount = (state.evilmCount || 0) + 1
    return `☐ 你选择虚开发票冲账。短期虚增成本少缴税，但虚开发票是刑事红线，随时可能爆雷。`
  }
  return `☐ 你选择了一条邪道。`
}
export function evilSalary(state) {
  const due = +(state.co.economics.salary * (state.scale || 1)).toFixed(1)
  state.evilmCount = (state.evilmCount || 0) + 1
  state.wageUnpaid = +((state.wageUnpaid || 0) + due).toFixed(2)
  state.balances['应付职工薪酬'] = +((state.balances['应付职工薪酬'] || 0) + due).toFixed(2)
  const bump = +(16 + Math.random() * 12 + (state.wageStrikes || 0) * 4).toFixed(1)
  state.wageDiscontent = Math.min(100, +((state.wageDiscontent || 0) + bump).toFixed(1))
  return `☐ 你拖欠了本月工资（欠薪累计 ☐${state.wageUnpaid}万）。员工满意度下滑，不满度升至 ${Math.round(state.wageDiscontent)}`
}
export function evilTaxAdjust(state, ratio = 0.5) {
  const due = +sumAccount(state.balances, '应交税费').toFixed(2)
  if (due <= 0) return `当前没有应纳税额可调整。`
  const cut = +(due * Math.min(0.9, Math.max(0, ratio))).toFixed(2)
  state.evilmCount = (state.evilmCount || 0) + 1
  state.taxAdjusts = (state.taxAdjusts || 0) + 1
  state.taxAdjustAmount = +((state.taxAdjustAmount || 0) + cut).toFixed(2)
  const entries = [
    ...mk('应交税费', cut, '所得税费用', cut, '调账：调减本期应纳税额（违规）'),
  ]
}

```

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```

  applyBusiness(state, entries, '税务调账（违规）', state.month)
  return `☐ 你通过调账把应纳税额调减了 ☐${cut}万（累计已调减 ☐${state.taxAdjustAmount}万）。账面少缴了税，但调账次数越多，`
}
export function evilTaxOwe(state) {
  const due = +sumAccount(state.balances, '应交税费').toFixed(2)
  if (due <= 0) return `当前没有应纳税额可拖欠。`
  state.evilmCount = (state.evilmCount || 0) + 1
  state.taxOwed = +((state.taxOwed || 0) + due).toFixed(2)
  state.skippedTaxMonths = (state.skippedTaxMonths || 0) + 1
  return `☐ 你直接拖欠了本期税款 ☐${due}万（累计拖欠 ☐${state.taxOwed}万）。从拖欠之日起按日加收万分之五滞纳金，越拖越贵。`
}
export function resolveStrike(state, option) {
  state.wageStrikes = (state.wageStrikes || 0) + 1 // 无论哪种都记一次黑历史
}

```

```

const owed = state.wageUnpaid || 0
const cash = state.balances['银行存款'] || 0
let pay, note
if (option === 'partial') {
  pay = +(owed * 0.5).toFixed(2)
  state.wageDiscontent = Math.max(0, +((state.wageDiscontent || 0) - 40).toFixed(1))
  state.wageUnpaid = +(owed - pay).toFixed(2)
  note = `部分补发：发放 □${pay}万（欠薪还剩 □${state.wageUnpaid}万），不满意度下降但仍未平复。`
} else if (option === 'over') {
  pay = +(owed * 1.5).toFixed(2)
  state.wageDiscontent = 0
  state.wageUnpaid = 0
  note = `超额补发：发放 □${pay}万（含 50% 安抚金），员工感激，不满意度清零、欠薪结清。`
} else { // full
  pay = +owed.toFixed(2)
  state.wageDiscontent = 0
  state.wageUnpaid = 0
  note = `全额补发：发放 □${pay}万，欠薪结清、不满意度清零，但这次罢工已记入黑历史。`
}
pay = Math.min(pay, +(cash + owed).toFixed(2)) // 不会凭空变出钱
const entries = [
  ...mk('应付职工薪酬', Math.min(owed, pay), '银行存款', Math.min(owed, pay), `罢工后补发工资(${option})`),
]
if (option === 'over' && pay > owed) {
  entries.push(...mk('管理费用-工资', +(pay - owed).toFixed(2), '银行存款', +(pay - owed).toFixed(2), `罢工安抚金(超额)`))
}
applyBusiness(state, entries, '罢工补发', state.month)
state.pendingStrike = null
state.strikeThisMonth = true
return `□ 罢工结束。${note}`
}
export function wageEfficiency(state) {
  const d = state.wageDiscontent || 0
  let f = 1 - (d / 100) * 0.45 // 不满意度最高拉低 45% 效率
  if (state.strikeThisMonth) f *= 0.4 // 罢工当月再打四折
  return +Math.max(0.25, f).toFixed(3)
}
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```

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```

export function accrueTaxLateFee(state) {
  if ((state.taxOwed || 0) <= 0) return 0
  const fee = +(state.taxOwed * 0.0005 * 30).toFixed(3)
  state.taxLateFee = +((state.taxLateFee || 0) + fee).toFixed(3)
  return fee
}
export function rollStrike(state) {
  state.strikeThisMonth = false
}

```

```

    if ((state.wageUnpaid || 0) <= 0) return false
    const d = state.wageDiscontent || 0
    const p = Math.min(0.85, 0.05 + (d / 100) * 0.45 + (state.wageStrikes || 0) * 0.08)
    if (Math.random() <= p) {
        state.strikeThisMonth = true
        state.pendingStrike = { month: state.month }
        return true
    }
    return false
}

export function taxAudit(state, month) {
    state.pendingAudit = null
    const adjustP = Math.min(0.6, 0.08 * (state.taxAdjusts || 0)) // 调账次数越多越易查
    const oweP = Math.min(0.5, 0.06 * (state.skippedTaxMonths || 0)) // 拖欠越久越易查
    const p = Math.max(adjustP, oweP)
    if (p <= 0) return null
    if (Math.random() > p) return null
    const year = state.year
    const events = []
    let entries = []
    state.auditHits = (state.auditHits || 0) + 1
    if ((state.taxAdjustAmount || 0) > 0) {
        const back = +state.taxAdjustAmount.toFixed(2)
        const fine = +(back * 0.5).toFixed(2)
        entries.push(...mk('所得税费用', back, '应交税费', back, `税务稽查：调账不实，补回税款□${back}万`))
        entries.push(...mk('所得税费用', fine, '银行存款', fine, `税务稽查罚款□${fine}万`))
        events.push(`调账被查：补回税款□${back}万、罚款□${fine}万`)
        state.taxAdjustAmount = 0 // 已补回，累计虚减额清零
    }
    if ((state.taxOwed || 0) > 0) {
        const back = +state.taxOwed.toFixed(2)
        const late = +(state.taxLateFee || 0).toFixed(2)
        entries.push(...mk('应交税费', back, '银行存款', back, `税务稽查：补缴拖欠税款□${back}万`))
        if (late > 0) entries.push(...mk('管理费用-滞纳金', late, '银行存款', late, `补缴滞纳金□${late}万`))
        events.push(`拖欠被查：补缴税款□${back}万${late > 0 ? `、滞纳金□${late}万` : ''}`)
        state.taxOwed = 0; state.taxLateFee = 0; state.skippedTaxMonths = 0
    }
    if (entries.length) {
        applyBusiness(state, entries, `税务稽查（第${year}年${month}月）`, month)
    }
    const msg = `□ 第${year}年${month}月税务稽查！${events.join('; ')}。`
    state.pendingAudit = msg
}
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```

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```

    if (isBankrupt(state)) { state.failed = true; state.failedReason = msg }
    return msg
}

```

```

export function maybeEvilBlowup(state, month) {
  state.pendingBlowup = null
  if ((state.evilCount || 0) === 0) return null
  if (month % 12 !== 0) return null
  const p = Math.min(0.85, 0.12 * (state.evilCount || 0)) // 单次爆雷概率随违规数上升
  if (Math.random() > p) return null
  const year = state.year
  const taxDue = +sumAccount(state.balances, '应交税费').toFixed(2)
  const wageDue = +state.balances['应付职工薪酬'].toFixed(2)
  const events = []
  let entries = []
  if (taxDue > 0) {
    const fine = +(taxDue * 0.5).toFixed(2)
    entries.push(...mk('所得税费用', fine, '应交税费-滞纳金', fine, `税务稽查：追缴税款□${taxDue}万 + 罚款□${fine}万`))
    entries.push(...mk('应交税费-滞纳金', fine, '银行存款', fine, '缴纳滞纳金罚款'))
    events.push(`税务稽查：追缴税款□${taxDue}万、罚款□${fine}万`)
  }
  if (wageDue > 0) {
    const fine = +(wageDue * 0.5).toFixed(2)
    entries.push(...mk('管理费用', fine, '银行存款', fine, `劳动稽查：补发欠薪□${wageDue}万 + 赔偿□${fine}万`))
    events.push(`劳动稽查：补发欠薪□${wageDue}万、赔偿□${fine}万`)
  }
  if (entries.length) {
    applyBusiness(state, entries, `邪道爆雷（第${year}年）`, month)
    state.skippedTaxMonths = 0
  }
  const msg = `□ 第${year}年爆雷! ${events.join('; ')}`。邪道终有代价。`
  state.pendingBlowup = msg
  if (isBankrupt(state)) {
    state.failed = true
    state.failedReason = msg
  }
  return msg
}

export function liabilityTotal(state) {
  return +sumAccount(state.balances, '短期借款') + sumAccount(state.balances, '应付账款') +
    sumAccount(state.balances, '应付职工薪酬') + sumAccount(state.balances, '应付利息') +
    sumAccount(state.balances, '应交税费')
}

export function operatingRevenue(state) {
  return +sumAccount(state.balances, '主营业务收入')
}

export function genOrder(state, rng = Math.random) {
  const co = state.co
  const base = co.economics.dealSize * (state.scale || 1)
  const sizeRoll = 0.6 + rng() * 0.9 // 0.6~1.5 倍基准
  const we = wageEfficiency(state)

```

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```

    }
    if (a.type === 'evilTaxOwe') {
      const msg = evilTaxOwe(s)
      setToast(msg)
      return { handled: true, selfContained: false }
    }
    if (a.type === 'resolveStrike') {
      const msg = resolveStrike(s, a.option)
      setToast(msg)
      return { handled: true, selfContained: false }
    }
    if (a.type === 'evilFakeInvoice') {
      const msg = evilAct(s, 'fakeInvoice')
      setToast(msg)
      return { handled: true, selfContained: false }
    }
    if (a.type === 'loopStart' || a.type === 'loopContinue') {
      if (a.type === 'loopStart') {
        s.year = 1
        s.yearMood = rollYearMood() // 第 1 年行情开局即定
        s.reachedMilestones = []
        s.choiceLog = s.choiceLog || []
        s.receivables = []
        s.payablesDue = []
      }
      if (chapter.id === 'continuing') {
        const orders = Array.from({ length: 3 }, () => genOrder(s))
        s.monthOrders = orders
        setMonthOrders(orders)
      } else {
        setMonthOrders([])
      }
      const back = 1 // 回到"进货"步 (step1), 重复 进→销→薪→结
      setSim(s); setStepIdx(back); persist(s, cold, diffId, chapterIdx, back, false)
      const orderCount = (s.monthOrders || []).length
      setToast(a.type === 'loopStart' ? `□ 开始持续经营! 本年行情: ${MOOD_LABEL[s.yearMood]} (第1年)` : `□ 进入第 ${s.mon`
      return { handled: true, selfContained: true }
    }
    if (a.type === 'close') {
      setSim(s)
      finish(s)
      return { handled: true, selfContained: true }
    }
    return { handled: false }
  }
  const choose = (opt) => {
    try {
      if (!opt || !opt.action) { nextStep(); return }

```

```

    const a = opt.action
    const before = clone(sim)

```

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```

    const s = clone(sim)
    const sp = runSpecial(s, a)
    if (sp.handled) {
        if (!sp.selfContained) {
            afterSpecial(s)
        } else {
            setAnalysis(analyzeDecision(before, s, a, { title: '操作被拦截' }))
        }
        return
    }
    const { entries, expected, desc } = doBusiness(s, a)
    if (diff.selfEntry) return // 困难模式不应走这里
    recordVoucher(s, { desc, actual: expected, expected, month: s.month })
    setSim(s); setLastEntries(entries); setToast(desc + ' □')
    setAnalysis(analyzeDecision(before, s, a))
    afterAction(s)
} catch (e) {
    console.error('[choose 出错]', e)
    setRuntimeError(`选择出错: ${e} && e.message\n${e} && e.stack`)
}
}

const submitEntry = () => {
    try {
        const a = step.options[0]?.action
        if (!a) { nextStep(); return }
        const before = clone(sim)
        const s = clone(sim)
        const sp = runSpecial(s, a)
        if (sp.handled) {
            if (!sp.selfContained) {
                afterSpecial(s)
            } else {
                setAnalysis(analyzeDecision(before, s, a, { title: '操作被拦截' }))
            }
            return
        }
        const { entries, expected, desc } = doBusiness(s, a, brief?.amount)
        const actual = [
            { side: 'debit', account: entryForm.dAcc, amount: +entryForm.dAmt },
            { side: 'credit', account: entryForm.cAcc, amount: +entryForm.cAmt },
        ]
        const v = recordVoucher(s, { desc, actual, expected, month: s.month })
        setSim(s)
    }
}

```

```

    if (v.correct) {
      setToast('☐ 分录正确！')
      setLastEntries(entries)
    } else {
      loseLife(s)
      setToast(diff.autoFix ? '记错啦，已自动修正' : `☐ 分录有误，扣 1 次容错（剩 ${s.lives}）`)
      setLastEntries(entries) // 展示正确分录

```

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```

    }
    setAnalysis(analyzeDecision(before, s, a, { title: '自写分录: ' + desc }))
    setEntryForm({ dAcc: '', dAmt: '', cAcc: '', cAmt: '' })
    afterAction(s)
  } catch (e) {
    console.error('[submitEntry 出错]', e)
    setRuntimeError(`提交分录出错: ${e && e.message}\n${e && e.stack}`)
  }
}
const afterSpecial = (s) => {
  if (isBankrupt(s)) { fail(s, '资不抵债，公司破产!'); return }
  if (s.failed) { fail(s, s.failedReason); return }
  setSim(s); persist(s, cold, diffId, chapterIdx, stepIdx, false)
  nextStep()
}
const afterAction = (s) => {
  if (isBankrupt(s)) { fail(s, '资不抵债，公司破产!'); return }
  if (s.failed) { fail(s, s.failedReason); return }
  nextStep()
}
const doMonthEnd = (passed) => {
  const before = clone(sim) // 结账前快照（用于月度体检对比）
  const s = passed || clone(sim)
  monthEnd(s)
  const extras = endOfMonthExtras(s, { rollEvent: true })
  if (extras.ev) setToast(`☐ ${extras.ev.emoji} ${extras.ev.title}: ${extras.ev.desc}`)
  if (extras.miles && extras.miles.length) {
    setTimeout(() => setToast(`☐ 里程碑达成: ${extras.miles.map((m) => m.emoji + m.title).join(', ')} `), 1200)
  }
  const settle = settleErrors(s)
  if (settle.fatal) { fail(s, s.failedReason); return }
  setSim(s)
  if (s.pendingStrike) {
    setPendingStrike({ month: s.month })
  }
  let msg = `第${s.month}月结账: 折旧+计息${settle.penalized ? `，罚款☐${settle.penalized}万` : ''}`
  if (extras.events && extras.events.length) {
    msg += `; ` + extras.events.map((e) => `${e.title}☐${fmtW(e.amount)}万`).join(', ')
  }

```

```

    }
    if (settle.tasks && settle.tasks.length) {
      setAdjustTasks(settle.tasks)
      msg += '； ☐ 本月有凭证记错，需做调整分录修正！ '
    }
    if (s.pendingBlowup) msg += `； ${s.pendingBlowup}`
    setToast(msg)
    if (s.pendingBlowup) s.pendingBlowup = null
    setAnalysis(analyzeDecision(before, s, { type: 'monthEnd' }, { title: `第 ${s.month} 月结账体检` }))
    if (s.failed) { fail(s, s.failedReason); return }
    if (isBankrupt(s)) { fail(s, '资不抵债，公司破产！'); return }
    nextStep()
  }
}
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```

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```

}
const submitAdjust = (task) => {
  const s = clone(sim)
  const entries = [
    { side: 'debit', account: entryForm.dAcc, amount: +entryForm.dAmt },
    { side: 'credit', account: entryForm.cAcc, amount: +entryForm.cAmt },
  ]
  applyAdjust(s, task.vid, entries)
  setAdjustTasks((t) => t.filter((x) => x.vid !== task.vid))
  setEntryForm({ dAcc: '', dAmt: '', cAcc: '', cAmt: '' })
  setToast(' ☐ 调整分录已应用')
  setSim(s)
}
const acceptOrder = (order) => {
  try {
    const s = clone(sim)
    fulfillOrder(s, order)
    s.monthOrders = (s.monthOrders || []).filter((o) => o.id !== order.id)
    setMonthOrders(s.monthOrders)
    setSim(s)
    persist(s, cold, diffId, chapterIdx, stepIdx, false)
    setToast(` ☐ 已接「${order.customer}」订单：销售额 ☐ ${fmtW(order.amount)} 万，预计毛利 ☐ ${fmtW(order.profit)} 万 ${order.customer}`)
  } catch (e) {
    console.error('[acceptOrder 出错]', e)
    setRuntimeError(`接单出错：${e && e.message}\n${e && e.stack}`)
  }
}
const nextStep = () => {
  if (stepIdx + 1 < chapter.steps.length) {
    const n = stepIdx + 1
    setStepIdx(n)
    persist(sim, cold, diffId, chapterIdx, n, false)
  } else if (chapterIdx + 1 < STORY.length) {

```

```

    const ci = chapterIdx + 1
    setChapterIdx(ci); setStepIdx(0)
    persist(sim, cold, diffId, ci, 0, false)
  } else {
    finish(sim)
  }
}
const finish = (s) => {
  const r = buildReports(s)
  setReports(r)
  const profit = s.balances['本年利润'] || 0
  const cash = s.balances['银行存款']
  const ok = cash >= 0 && !s.failed
  const scores = scoreMetrics(s)
  const stars = overallStars(scores)
  const res = {
    ok, stars,
  }
}
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```

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```

    reason: ok ? `通关！净利润${fmtW(profit)}万 现金${fmtW(cash)}万` : '未达成通关条件',
    profit, cash, scores,
    insights: decisionInsights(s), // B: 决策后果回放
    milestones: (s.reachedMilestones || []).map((id) => MILESTONES.find((m) => m.id === id)).filter(Boolean), // D
    ordersFulfilled: s.choices?.ordersFulfilled || 0,
  }
  setResult(res)
  setEnded(true)
  dispatch({ type: 'ADD_COINS', amount: ok ? 50 : 20 })
  dispatch({ type: 'ADD_EXP', amount: 60 })
  dispatch({ type: 'COMPANY_RUN', id: cold })
  dispatch({ type: 'EARN_BADGE', id: ok ? 'boss' : 'cfo' })
  dispatch({
    type: 'RECORD_RUN',
    cold, coName: sim.co?.name || cold,
    profit: +(sim.balances['本年利润'] || 0).toFixed(2),
    stars: res.stars, months: sim.month, date: new Date().toISOString().slice(0, 10),
  })
  res.milestones.forEach((m) => dispatch({ type: 'EARN_BADGE', id: 'ms_' + m.id }))
  clearSave(); setSaved(null)
}
const fail = (s, reason) => {
  setSim(s)
  const res = { ok: false, stars: 0, reason, profit: s.balances['本年利润'] || 0, cash: s.balances['银行存款'] }
  setResult(res)
  setEnded(true)
  dispatch({ type: 'ADD_COINS', amount: 10 })
  persist(s, cold, diffId, chapterIdx, stepIdx, true)
}

```

```

    setResult((r) => ({ ...r, failed: true }))
  }
  const restart = () => {
    clearSave(); setSaved(null)
    setPhase('select'); setSim(null); setReports(null); setEnded(false); setResult(null); setAdjustTasks([]); setStepIdx(0)
  }
  const clone = (s) => JSON.parse(JSON.stringify(s))
  function mk(d, da, c, ca, desc) {
    return [
      { side: 'debit', account: d, amount: da, desc },
      { side: 'credit', account: c, amount: ca, desc },
    ]
  }
}
if (phase === 'askLoad' && saved) {
  const savedCo = COMPANIES.find((c) => c.id === saved.cold)
  const savedDiff = DIFFICULTY[saved.diffId]
  return (
    <div className="page fade-in">
      <div className="card" style={{ textAlign: 'center', marginTop: 40 }}>
        <div style={{ fontSize: 46 }}>□</div>
        <div style={{ fontWeight: 800, fontSize: 18, marginTop: 8 }}>检测到之前的经营存档</div>
        <div style={{ color: 'var(--text-soft)', marginTop: 6 }}>
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```

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```

    {savedCo?.emoji} {savedCo?.name} · {savedDiff?.name}
  </div>
  <div style={{ color: 'var(--text-soft)', fontSize: 12, marginTop: 2 }}>
    进度: 第 {saved.chapterIdx + 1} 章第 {saved.stepIdx + 1} 步
  </div>
  <button className="btn mt12" onClick={() => resume(saved)}>□ 继续上次的经营</button>
  <button className="btn ghost mt12" onClick={() => { clearSave(); setSaved(null); setPhase('select') }}>□ 重新开一
</div>
</div>
)
}
if (phase === 'select') {
  if (pendingCompany) {
    const c = pendingCompany
    const projects = c.projects || []
    return (
      <div className="page fade-in">
        <Toast message={toast} onClose={() => setToast('')} />
        <button className="btn ghost" style={{ marginBottom: 10 }} onClick={() => setPendingCompany(null)}>← 换家公司</button>
        <div className="section-title">□ {c.emoji} {c.name} · 选经营项目</div>
        <div style={{ color: 'var(--text-soft)', fontSize: 12, marginBottom: 8 }}>
          同一类公司也分不同产品线, 参数与难点略有差异, 挑一个你熟悉的吧~
        </div>

```

```

    {projects.length === 0 ? (
      <div className="card">该产品线暂未细分经营项目，直接选择难度开始。</div>
    ) : projects.map((p) => (
      <div key={p.id} className="card">
        <div style={{ fontWeight: 800 }}>{p.emoji} {p.name}</div>
        <div style={{ color: 'var(--text-soft)', fontSize: 12, marginTop: 4 }}>{p.blurb}</div>
        {p.econ && (
          <div className="flex" style={{ marginTop: 6 }}>
            {p.econ.margin !== null && <span className="chip">毛利 {Math.round(p.econ.margin * 100)}%</span>}
            {p.econ.dealSize !== null && <span className="chip">单笔 ☐ {p.econ.dealSize} 万</span>}
          </div>
        )}
      <div className="section-title" style={{ margin: '12px 0 8px', fontSize: 14 }}>选择难度</div>
      {Object.values(DIFFICULTY).map((d) => (
        <button key={d.id} className="btn ghost mt12" style={{ width: '100%', padding: '10px' }} onClick={() => start(
          {d.name} · 容错 {d.lives} 次 {d.selfEntry ? ' · 自写分录' : ' · 选项引导'}
        )}>
      </button>
    ))}
    </div>
  ))}
  {projects.length > 0 && (
    <button className="btn mt12" style={{ width: '100%' }} onClick={() => start(c.id, diffId, projects[0].id)}>
      直接开干（默认 {projects[0].emoji} {projects[0].name} · {DIFFICULTY[diffId].name}
    </button>
  )}
</div>
)

```

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```

}
return (
  <div className="page fade-in">
    <Toast message={toast} onClose={() => setToast('')} />
    <div className="section-title"><input type="checkbox"/> 选择公司类型</div>
    {COMPANIES.map((c) => (
      <div key={c.id} className="card" style={{ cursor: 'pointer' }} onClick={() => chooseCompany(c.id)}>
        <div style={{ fontWeight: 800 }}>{c.emoji} {c.name}</div>
        <div style={{ color: 'var(--text-soft)', fontSize: 12, marginTop: 4 }}>{c.blurb}</div>
        <div className="flex" style={{ marginTop: 6 }}>
          <span className="chip">难度 {'★'.repeat(c.difficulty)}</span>
          <span className="chip">启动 ☐ {c.initCash} 万</span>
          {c.projects?.length > 0 && <span className="chip">{c.projects.length} 个经营项目</span>}
        </div>
      </div>
    ))}
  </div>
)

```



```

    </div>
  </div>
)
}
if (adjustTasks.length > 0) {
  const task = adjustTasks[0]
  return (
    <div className="page fade-in">
      <Toast message={toast} onClose={() => setToast('')} />
      <div className="card" style={{ borderLeft: '5px solid var(--gold)' }}>
        <div style={{ fontWeight: 800, color: 'var(--gold)' }}>□ 调整分录任务</div>
        <div style={{ marginTop: 6 }}>业务: {task.desc}</div>
        <div style={{ fontSize: 12, color: 'var(--text-soft)', marginTop: 6 }}>你记成了: {task.wrong.map((e) => `${e.side`
        <div style={{ fontSize: 12, color: 'var(--accent-deep)', background: '#E9F8F6', padding: 8, borderRadius: 8, marg
        <EntryForm form={entryForm} setForm={setEntryForm} />
        <button className="btn mt12" onClick={() => submitAdjust(task)}>提交调整分录</button>
      </div>
    </div>
  )
}
const selfMode = diff.selfEntry && step?.options?.[0]?.action
const entryActionTypes = ['rent', 'fixed', 'purchase', 'purchaseCredit', 'sale', 'salary']
const curAction = step?.options?.[0]?.action
const isEntryAction = curAction && entryActionTypes.includes(curAction.type)
if (pendingStrike) {
  const owed = sim.wageUnpaid || 0
  const d = Math.round(sim.wageDiscontent || 0)
  const resolve = (option) => {
    runSpecial({ type: 'resolveStrike', option })
    setPendingStrike(null)
  }
}
return (
  <div className="page fade-in">
    <Toast message={toast} onClose={() => setToast('')} />
    <div className="card" style={{ borderLeft: '5px solid #e74c3c', background: '#FFF5F5', marginTop: 30 }}>
      <div style={{ fontSize: 30 }}>□</div>
      <div style={{ fontWeight: 800, color: '#e74c3c', marginTop: 4 }}>员工罢工了! </div>

```

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```

<div style={{ marginTop: 8, fontSize: 13, lineHeight: 1.7 }}>
  因长期拖欠工资, 员工发起罢工。当前累计欠薪 <b>□ {fmtW(owed)} 万</b>, 不满度 <b>{d}/100</b>。
  罢工当月效率骤降。你决定如何回应? (无论哪种都会记入罢工黑历史, 使下次更易罢工、损失更大)
</div>
<div style={{ display: 'flex', flexDirection: 'column', gap: 8, marginTop: 14 }}>
  <button className="btn" onClick={() => resolve('full')}>□ 全额补发 (结清欠薪, 不满度清零)</button>
  <button className="btn ghost" onClick={() => resolve('partial')}>□ 部分补发 50% (欠薪减半, 不满度下降但仍未平复
  <button className="btn ghost" onClick={() => resolve('over')}>□ 超额补发 150% (多花 50% 安抚金, 彻底挽回人心)</button>

```

```

    </div>
  </div>
</div>
)
}
return (
  <div className="page fade-in">
    {runtimeError && (
      <div className="card" style={{ borderLeft: '5px solid #e74c3c', background: '#FFF5F5', marginBottom: 12 }}>
        <div style={{ fontWeight: 800, color: '#e74c3c', fontSize: 15 }}>❌ 交互出错</div>
        <pre style={{ whiteSpace: 'pre-wrap', fontSize: 12, background: '#fbeaea', padding: 10, borderRadius: 8, marginTop: 10 }}>{runtimeError}</pre>
        <button className="btn mt12" onClick={() => setRuntimeError(null)}>关闭提示</button>
      </div>
    )}
    <Toast message={toast} onClose={() => setToast('')} />
    <div className="card">
      <div className="flex between center">
        <div style={{ fontWeight: 800 }}>[co.emoji] [co.name] [sim.projectEmoji ? ` · ${sim.projectEmoji}` : ''][sim.projectName]</div>
        <span className="chip">[diff.name]</span>
      </div>
      <div className="flex gap8 mt12">
        <div className="shop-stat" style={{ flex: 1, marginBottom: 0 }}><span>💰 现金</span><span className="v">[{fmtW(s.shop.cash)}]</span></div>
        <div className="shop-stat" style={{ flex: 1, marginBottom: 0 }}><span>📈 利润</span><span className="v">[{fmtW(s.shop.profit)}]</span></div>
        <div className="shop-stat" style={{ flex: 1, marginBottom: 0 }}><span>⚠️ 容错</span><span className="v">[sim.lives]</span></div>
      </div>
      <div className="flex gap8 mt8">
        <div className="shop-stat" style={{ flex: 1, marginBottom: 0 }}><span>💵 营收</span><span className="v">[{fmtW(s.shop.revenue)}]</span></div>
        <div className="shop-stat" style={{ flex: 1, marginBottom: 0 }}><span>💳 负债</span><span className="v">[{fmtW(s.shop.debt)}]</span></div>
        <div className="shop-stat" style={{ flex: 1, marginBottom: 0 }}><span>🏢 规模</span><span className="v">× [{fmtW(s.shop.scale)}]</span></div>
      </div>
      {(chapter.loop) && (
        <div className="flex gap8 mt8">
          <div className="shop-stat" style={{ flex: 1, marginBottom: 0 }}><span>📊 行情</span><span className="v">[MOOD]</span></div>
          <div className="shop-stat" style={{ flex: 1, marginBottom: 0 }}><span>👹 邪道</span><span className="v">[sim.evil]</span></div>
        </div>
        /* C: 当前随机事件提示条 */
        {sim.lastEvent && (
          <div className="flex gap8 mt8" style={{ alignItems: 'center' }}>
            <div className="shop-stat" style={{ flex: 1, marginBottom: 0, background: sim.lastEvent.tone === 'good' ? '#d4edda' : '#f8d7da' }}>
              <span>[sim.lastEvent.emoji] [sim.lastEvent.title]</span>
              <span className="v" style={{ fontSize: 11, fontWeight: 500 }}>[sim.lastEvent.desc]</span>
            </div>
          </div>
        )}
      )}
    </div>
  )
}
```

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```

    </div>
  </div>
)}

```

```

    /* D: 里程碑进度 */
    [MILESTONES.length > 0 && (
      <div className="flex gap8 mt8" style={{ flexWrap: 'wrap' }}>
        {MILESTONES.map((m) => {
          const done = (sim.reachedMilestones || []).includes(m.id)
          const near = sim.month >= m.month - 1 && !done
          return (
            <span key={m.id} title={m.desc}
              style={{ fontSize: 11, padding: '3px 8px', borderRadius: 8, opacity: done ? 1 : 0.4,
                background: done ? 'var(--gold)' : near ? '#E9F8F6' : '#f2efe8', color: done ? '#fff' : 'var(--text-soft)',
                {done ? '□' : near ? '□' : '□'} {m.title}
              </span>
            )
          )
        })}
      </div>
    )
  }
  </>
)
}
{<chapter.loop> && (sim.totalDividend > 0 || (sim.usedTaxPlans || []).length) && (
  <div className="flex gap8 mt8">
    <div className="shop-stat" style={{ flex: 1, marginBottom: 0 }}><span>□ 累计分红</span><span className="v">□ {t
    <div className="shop-stat" style={{ flex: 1, marginBottom: 0 }}><span>□ 合法筹划</span><span className="v">{(s
    </div>
  )}
</div>
<div className="card" style={{ background: '#FFFDF8', borderLeft: '5px solid var(--primary)' }}>
  <div style={{ fontWeight: 700, color: 'var(--primary-deep)', fontSize: 13 }}>
    {chapter.title} · 第{sim.month}月{chapter.loop ? ` (第 ${Math.ceil(sim.month / 12)} 年)` : ''}
  </div>
  <div style={{ marginTop: 6, lineHeight: 1.7 }}>{step?.npc}</div>
  {step?.teach && (
    <div style={{ marginTop: 8, fontSize: 12, color: 'var(--accent-deep)', background: '#E9F8F6', padding: '8px 10px',
    )}
    <button className="btn ghost mt12" style={{ width: 'auto', padding: '6px 12px', fontSize: 12 }} onClick={() => setS
      {showHint ? '收起提示' : '□ 卡住了? 看思路'}
    </button>
    {showHint && (step?.teach || (step?.options || []).some((o) => o.teach)) && (
      <div style={{ fontSize: 12, color: 'var(--text-soft)', marginTop: 6, lineHeight: 1.7 }}>
        {step?.teach && <div>思路: {step.teach} (想想这笔业务影响了哪些科目、是增加还是减少、记借还是贷)</div>}
        {(step?.options || []).filter((o) => o.teach).map((o, i) => (
          <div key={i} style={{ marginTop: step?.teach ? 6 : 0 }}>
            {step?.teach ? `□ ${o.label}: ` : '思路: '} {o.teach}
          </div>
        ))}
      </div>
    )}
  </div>
)
}
</div>

```

```

    /* E: 持续经营"销售"步——主动接单面板（替换被动卖货） */
    {!selfMode && chapter.id === 'continuing' && step?.options?.[0]?.action?.type === 'sale' && (
      <div className="card">
        <div style={{ fontWeight: 700, marginBottom: 4 }}>□ 本月订单（点击接单，把生意主动权握在手里）</div>
        <div style={{ fontSize: 12, color: 'var(--text-soft)', marginBottom: 10 }}>
          下方订单可全接也可挑着接。赊销订单会挂「应收账款」，数月后自动回款；不接单也可直接点"卖货收款"按默认成交一笔。
        </div>
        {monthOrders.length === 0 ? (
          <div style={{ fontSize: 13, color: 'var(--text-soft)', padding: '8px 0' }}>本月订单都已接完，或本月无新订单。</div>
        ) : (
          <div style={{ display: 'grid', gap: 10 }}>
            {monthOrders.map((o) => (
              <div key={o.id} style={{ border: '1.5px solid var(--line)', borderRadius: 12, padding: '10px 12px', background: 'var(--card)' }}>
                <div style={{ fontWeight: 700, fontSize: 13 }}>{o.emoji || '□'} {o.customer}</div>
                <div style={{ fontSize: 12, color: 'var(--text-soft)', margin: '4px 0' }}>
                  销售额 □{fmtW(o.amount)}万 · 毛利率 {Math.round(o.margin * 100)}% · 预计毛利 □{fmtW(o.profit)}万
                </div>
                <div style={{ fontSize: 11, marginBottom: 6 }}>
                  {o.credit > 0
                    ? <span style={{ color: 'var(--accent-deep)', background: '#E9F8F6', padding: '2px 8px', borderRadius: 6 }}>现结
                    : <span style={{ color: '#1a8f6a', background: '#E3F6EF', padding: '2px 8px', borderRadius: 6 }}>现结
                  </div>
                  <button className="btn-sm" onClick={() => acceptOrder(o)} style={{ width: '100%' }}>接单并记账</button>
                </div>
              </div>
            ))}
          </div>
        )}
      </div>
    )}
  </div>
  /* 简单模式：选项卡片 */
  {!selfMode && (
    <div className="card">
      {step?.options?.length ? (
        step.options.map((opt, i) => (
          <div key={i}>
            <div className="option" style={{ cursor: 'pointer' }} onClick={() => choose(opt)}>
              <span className="opt-key">{i + 1}</span><span>{opt.label}</span>
              {opt.recommended && !diff.selfEntry && <span className="chip" style={{ marginLeft: 'auto', background: 'var(--chip)' }}>推荐
            </div>
            {diff.selfEntry === false && opt.demoEntries && (
              <div style={{ fontSize: 11, color: 'var(--accent-deep)', background: '#E9F8F6', padding: '6px 10px', border: '1px solid var(--border)' }}>
                □ 演示分录: {opt.demoEntries.map((e, k) => `${e.side === 'debit' ? '借' : '贷'} ${e.account} □${e.amount}`)}
              </div>
            )}
          </div>
        ))}
      </div>
    )}
  ) : (
    (step?.action || step?.options?.[0]?.action) ? (

```

```

    <div className="card" style={{ padding: 12 }}>
      <div style={{ fontWeight: 700, marginBottom: 8 }}>请填写这笔业务的分录:</div>

```

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```

      <EntryForm form={entryForm} setForm={setEntryForm} />
      <button className="btn mt12" onClick={submitEntry}>提交分录</button>
    </div>
  ) : <button className="btn" onClick={() => nextStep()}>继续 </button>
  )}
</div>
)}
{/* 困难/硬核: 自写分录 / 特殊决策确认 */}
{selfMode && (
  <div className="card">
    {isEntryAction ? (
      <>
        <div style={{ fontWeight: 700, marginBottom: 8 }}><input type="checkbox"/> 自己写出这笔业务的分录</div>
        {brief && (
          <div style={{ fontSize: 12, background: '#E9F8F6', color: 'var(--accent-deep)', padding: '8px 10px', border: 1px solid 'var(--text-soft)'}>
            <input type="checkbox"/> 业务简报: {brief.type === 'sale' && `卖出货物 <input type="checkbox"/> ${brief.amount} 万, 税率 ${ (brief.rate * 100) }% → 销项税 <input type="checkbox"/> ${brief.vat} 万` }
            {brief.type === 'purchase' && `现金进货 <input type="checkbox"/> ${brief.amount} 万${brief.vat ? `, 进项税 <input type="checkbox"/> ${brief.vat} 万 (可抵: <input type="checkbox"/> ${brief.vat} 万` }
            {brief.type === 'purchaseCredit' && `赊购进货 <input type="checkbox"/> ${brief.amount} 万${brief.vat ? `, 进项税 <input type="checkbox"/> ${brief.vat} 万` }
            {brief.type === 'salary' && `计提工资 <input type="checkbox"/> ${brief.amount} 万`}
            {brief.type === 'rent' && `房租 <input type="checkbox"/> ${brief.amount} 万 (含押金) `}
          </div>
        )}
      </div>
    ) : (
      <EntryForm form={entryForm} setForm={setEntryForm} />
      <button className="btn mt12" onClick={submitEntry}>提交分录</button>
    </div>
  ) : (
    step.options.length ? (
      <>
        <div style={{ fontWeight: 700, marginBottom: 8 }}><input type="checkbox"/> 这个决策由你拍板</div>
        <div style={{ fontSize: 12, color: 'var(--text-soft)', marginBottom: 10 }}>{typeof step.npc === 'function' ? step.npc() : 'NPC 决策'}</div>
        {step.options.map((opt, i) => (
          <div key={i} className="option" style={{ cursor: 'pointer' }} onClick={() => choose(opt)}>
            <span className="opt-key">{i + 1}</span><span>{opt.label}</span>
          </div>
        ))}
      </div>
    ) : <button className="btn" onClick={() => nextStep()}>继续 </button>
  )}
</div>
)}
{lastEntries && <EntryAnimation entries={lastEntries} />}
{analysis && (
  <div className="card" style={{ background: analysis.alert ? '#FFF6F4' : '#F7FBF8', border: `1px solid ${analysis.alert ? 'var(--text-soft)' : 'var(--text-soft)'}` }}>

```

```

<div style={{ display: 'flex', alignItems: 'center', justifyContent: 'space-between', marginBottom: 6 }}>
  <div style={{ fontWeight: 800 }}>□ 财务诊断 · {analysis.title}</div>
  <div style={{ fontSize: 11, color: 'var(--text-soft)' }}>本步决策影响</div>
</div>
{analysis.impacts.length > 0 && (
  <div style={{ display: 'flex', flexWrap: 'wrap', gap: 6, marginBottom: 8 }}>
    {analysis.impacts.map((it, i) => (

```

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```

    <span key={i} style={{
      fontSize: 12, fontWeight: 700, padding: '3px 8px', borderRadius: 8,
      background: it.good ? '#E6F6EC' : '#FDEAE6',
      color: it.good ? '#1F8A4C' : '#C0392B',
    }}>
      {it.label} {it.dir === 'up' ? '▲' : '▼'} {fmtW(it.value)}
    </span>
  ))}
</div>
)}
{analysis.focus && (
  <div style={{ fontSize: 12.5, lineHeight: 1.6, color: 'var(--text)', marginBottom: 6 }}>
    {analysis.focus}
  </div>
)}
<div style={{
  fontSize: 12.5, fontWeight: 700, lineHeight: 1.6, marginBottom: 8,
  color: analysis.alert ? '#C0392B' : '#1F8A4C',
}}>
  {analysis.diagnosis}
</div>
<div style={{ display: 'grid', gridTemplateColumns: 'repeat(auto-fit, minmax(110px, 1fr))', gap: 6 }}>
  [[
    { k: '现金(万)', v: fmtW(analysis.snapshot.cash), warn: analysis.snapshot.cash < 2 },
    { k: '累计净利(万)', v: fmtW(analysis.snapshot.netProfit), warn: analysis.snapshot.netProfit < 0 },
    { k: '负债率', v: analysis.snapshot.debtRatio + '%', warn: analysis.snapshot.debtRatio > 70 },
    { k: '合规度', v: analysis.snapshot.compliance + '/100', warn: analysis.snapshot.compliance < 40 },
  ].map((m, i) => (
    <div key={i} style={{
      background: '#fff', borderRadius: 8, padding: '6px 8px', textAlign: 'center',
      border: `1px solid ${m.warn ? '#F3C0B5' : '#ECECEC'}`,
    }}>
      <div style={{ fontSize: 10.5, color: 'var(--text-soft)' }}>{m.k}</div>
      <div style={{ fontSize: 15, fontWeight: 800, color: m.warn ? '#C0392B' : 'var(--text)' }}>{m.v}</div>
    </div>
  )))
</div>
</div>

```

```

    })
    {reports && (
      <div ref={reportRef} className="card" style={{ background: '#FFDF8' }}>
        <div style={{ fontWeight: 800, marginBottom: 8 }}><input type="checkbox"/> 当前报表（实时）</div>
        <div style={{ fontSize: 12, fontWeight: 700, color: 'var(--accent-deep)', marginBottom: 8 }}>
          <input type="checkbox"/> 报表已生成，可对照检查借贷逻辑（资产=负债+权益 / 收入-费用=利润）
        </div>
        <ReportLite report={reports.balance} />
        <ReportLite report={reports.income} />
        <ReportLite report={reports.cashflow} />
      </div>
    )}
  )}

```

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```

<div className="section-title mt20"><input type="checkbox"/> 账本（最新 6 笔）</div>
[...sim.ledger].slice(-6).reverse().map((e, i) => (
  <div key={i} className="card" style={{ padding: 10, fontSize: 12 }}>
    <div style={{ fontWeight: 600 }}>第{e.month}月 · {e.desc}</div>
    {e.debit && <div style={{ color: 'var(--primary-deep)' }}>借 {e.debit} <input type="checkbox"/>{e.amt}万</div>}
    {e.credit && <div style={{ color: 'var(--accent-deep)' }}>贷 {e.credit} <input type="checkbox"/>{e.amt}万</div>}
  </div>
)}}
<button className="btn ghost mt12" onClick={restart}><input type="checkbox"/> 重开一家</button>
<button className="btn ghost mt12" onClick={() => {
  try {
    const r = buildReports(sim)
    setReports(r)
    setReportReady(true)
    setTimeout(() => reportRef.current?.scrollIntoView({ behavior: 'smooth', block: 'start' }), 50)
  } catch (err) {
    setRuntimeError(err)
  }
}}><input type="checkbox"/> {reportReady ? '重新生成报表' : '生成报表'}</button>
</div>
)
}
export default function GameWithBoundary(props) {
  return (
    <ErrorBoundary>
      <Game {...props} />
    </ErrorBoundary>
  )
}
function EntryForm({ form, setForm }) {
  return (
    <div>
      <div className="flex gap8" style={{ marginBottom: 8 }}>

```

```

    <span style={{ fontWeight: 700, color: 'var(--primary-deep)', width: 30 }}>借</span>
    <select value={form.dAcc} onChange={(e) => setForm({ ...form, dAcc: e.target.value })} style={{ flex: 1, padding: 8,
      <option value="">选择科目</option>{ALL_ACCOUNTS.map((a) => <option key={a} value={a}>{a}</option>)}
    </select>
    <input type="number" value={form.dAmt} placeholder="金额" onChange={(e) => setForm({ ...form, dAmt: e.target.value })}
  </div>
  <div className="flex gap8">
    <span style={{ fontWeight: 700, color: 'var(--accent-deep)', width: 30 }}>贷</span>
    <select value={form.cAcc} onChange={(e) => setForm({ ...form, cAcc: e.target.value })} style={{ flex: 1, padding: 8,
      <option value="">选择科目</option>{ALL_ACCOUNTS.map((a) => <option key={a} value={a}>{a}</option>)}
    </select>
    <input type="number" value={form.cAmt} placeholder="金额" onChange={(e) => setForm({ ...form, cAmt: e.target.value })}
  </div>
  <div style={{ fontSize: 11, color: 'var(--text-soft)', marginTop: 6 }}>借贷金额需相等</div>
</div>
)
}
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```

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```

function ReportLite({ report }) {
  const title = report.type === 'balance' ? '资产负债表' : report.type === 'income' ? '利润表' : '现金流量表'
  return (
    <div style={{ marginBottom: 10 }}>
      <div style={{ fontWeight: 700, fontSize: 13, marginBottom: 4 }}>{title}</div>
      {report.type === 'cashflow' && report.note && (
        <div style={{ fontSize: 10.5, color: 'var(--text-soft)', marginBottom: 4 }}>({report.note})</div>
      )}
      {report.rows.map((r, i) => {
        if (r.level === 'h') return (
          <div key={i} style={{ fontSize: 12, fontWeight: 800, marginTop: 6, color: 'var(--accent-deep)' }}>{r.item}</div>
        )
        const isTotal = r.level === 'total' || r.emphasize
        const isSub = r.level === 'sub'
        const neg = r.value != null && r.value < 0
        return (
          <div key={i} className="flex between center" style={{
            fontSize: isTotal ? 12.5 : 12,
            fontWeight: isTotal ? 800 : (isSub ? 700 : 400),
            padding: '2px 0',
            paddingLeft: isSub || isTotal ? 8 : 0,
            borderTop: isTotal ? '1px solid #d9d2c4' : 'none',
            color: neg ? '#00392B' : (isTotal ? 'var(--primary-deep)' : 'var(--text)'),
          }}>
            <span>{r.item}</span>
            <span>{r.value == null ? '' : '□' + fmtW(r.value) + '万'}</span>
          </div>
        )
      })}
    </div>
  )
}

```

```

    )}
  </div>
)
}
function RadarChart({ scores }) {
  const axes = [
    { key: 'profit', label: '盈利能力', value: scores.profit },
    { key: 'cash', label: '现金流', value: scores.cash },
    { key: 'risk', label: '低风险', value: scores.risk },
    { key: 'compliance', label: '合规度', value: scores.compliance },
  ]
  const cx = 90, cy = 90, R = 66
  const n = axes.length
  const pt = (i, r) => {
    const ang = (Math.PI * 2 * i) / n - Math.PI / 2
    return [cx + r * Math.cos(ang), cy + r * Math.sin(ang)]
  }
  const poly = axes.map((a, i) => pt(i, (a.value / 100) * R).join(',')).join(' ')
  return (
    <div style={{ marginTop: 12 }}>
      <div style={{ fontWeight: 700, fontSize: 13, marginBottom: 4 }}>□ 经营能力雷达</div>
      <svg width="180" height="180" viewBox="0 0 180 180" style={{ maxWidth: 220 }}>
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```

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```

    [[0.25, 0.5, 0.75, 1].map((g, i) => (
      <polygon key={i} points={axes.map((_, k) => pt(k, R * g).join(',')).join(' ')} fill="none" stroke="#e6e1d6" />
    ))]
    {axes.map((a, i) => {
      const [x, y] = pt(i, R)
      return <line key={i} x1={cx} y1={cy} x2={x} y2={y} stroke="#e6e1d6" />
    })}
    <polygon points={poly} fill="rgba(46, 138, 108, 0.28)" stroke="var(--primary-deep)" strokeWidth="2" />
    {axes.map((a, i) => {
      const [x, y] = pt(i, (a.value / 100) * R)
      return <circle key={i} cx={x} cy={y} r="3" fill="var(--primary-deep)" />
    })}
    {axes.map((a, i) => {
      const [x, y] = pt(i, R + 14)
      return <text key={i} x={x} y={y} fontSize="10" textAnchor="middle" fill="var(--text-soft)">{a.label}</text>
    })}
  </svg>
  <div style={{ display: 'flex', flexWrap: 'wrap', gap: 6, justifyContent: 'center', marginTop: 4 }}>
    {axes.map((a) => (
      <span key={a.key} className="chip" style={{ fontSize: 11 }}>{a.label} {a.value}</span>
    ))}
  </div>
</div>

```

```

    )
  }
  /* ===== 文件: pages/Home.jsx ===== */
  import { useNavigate } from 'react-router-dom'
  import { useUser } from '../store/UserContext.jsx'
  import { COMPANIES } from '../data/companies.js'
  import { TIERS } from '../data/tiers.js'
  export default function Home() {
    const navigate = useNavigate()
    const { state } = useUser()
    const badgePct = Math.round((state.badges.length / 7) * 100)
    return (
      <div className="page fade-in">
        <div className="hero">
          <h1>嗨, {state.name}! □</h1>
          <p>开一家公司, 在经营中自然学会会计~</p>
          <div style={{ marginTop: 14 }}>
            <div className="progress-track" style={{ background: 'rgba(255, 255, 255, .35)' }}>
              <div className="progress-fill" style={{ width: badgePct + '%', background: '#fff' }} />
            </div>
            <div style={{ fontSize: 12, marginTop: 6, opacity: .95 }}>
              已获徽章 {state.badges.length}/7 · Lv. {state.level} · □{state.coins}
            </div>
          </div>
        </div>
        <div className="section-title">□ 开始经营</div>
        <div className="card" onClick={() => navigate('/game')} style={{ cursor: 'pointer', background: 'linear-gradient(135deg, #f0f0f0, #fff)' }}>

```

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```

    <div className="flex between center">
      <div>
        <div style={{ fontWeight: 800, fontSize: 16 }}>□ 经营我的公司</div>
        <div style={{ color: 'var(--text-soft)', fontSize: 13, marginTop: 4 }}>
          选公司类型与难度, 从创立到报表全过程实战
        </div>
      </div>
      <span style={{ fontSize: 22 }}>□</span>
    </div>
  </div>
  <div className="card" onClick={() => navigate('/reports')} style={{ cursor: 'pointer' }}>
    <div className="flex between center">
      <div>
        <div style={{ fontWeight: 800, fontSize: 16 }}>□ 财务报表中心</div>
        <div style={{ color: 'var(--text-soft)', fontSize: 13, marginTop: 4 }}>
          查看资产负债表 / 利润表 / 现金流量表
        </div>
      </div>
    </div>

```

```

    <span style={{ fontSize: 22 }}>□</span>
  </div>
</div>
<div className="section-title mt20">□ 选哪种公司练手? </div>
[COMPANIES.map((c) => (
  <div key={c.id} className="card" onClick={() => navigate('/game')} style={{ cursor: 'pointer' }}>
    <div className="flex between center">
      <div>
        <div style={{ fontWeight: 800 }}>{c.emoji} {c.name}</div>
        <div style={{ color: 'var(--text-soft)', fontSize: 12, marginTop: 3 }}>{c.blurb}</div>
      </div>
      <span className="chip">{'★'.repeat(c.difficulty)}</span>
    </div>
  </div>
))}
<div className="section-title mt20">□ 会计职称路径</div>
[TIERS.map((t) => (
  <div key={t.id} className="card" onClick={() => navigate('/game')} style={{ cursor: 'pointer' }}>
    <div className="flex between center">
      <div>
        <div style={{ fontWeight: 800 }}>{t.name}</div>
        <div style={{ color: 'var(--text-soft)', fontSize: 12, marginTop: 3 }}>{t.desc}</div>
      </div>
      <span style={{ fontWeight: 800, color: t.color }}>□</span>
    </div>
  </div>
))}
</div>
)
}
/* ===== 文件: pages/Me.jsx ===== */
import { useState } from 'react'
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```

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```

import { useUser } from '../store/UserContext.jsx'
import { BADGES } from '../data/badges.js'
import { TIERS } from '../data/tiers.js'
import { useNavigate } from 'react-router-dom'
function runScore(r) {
  return Math.round((r.profit || 0) * 10 + (r.stars || 0) * 50 + (r.months || 0) * 3)
}
const RANK_BASE = [
  { name: '会计学霸喵', score: 1820, avatar: '□' },
  { name: '算盘小能手', score: 1560, avatar: '□' },
  { name: '账房先生', score: 1340, avatar: '□' },
  { name: '小数点猎人', score: 980, avatar: '□' },
  { name: '借贷平衡侠', score: 760, avatar: '□' },

```

```

]
const CO_NAMES = {
  restaurant: '☐ 餐饮店', shop: '☐ 零售店', supermarket: '☐ 超市',
  factory: '☐ 工厂', tech: '☐ 科技工作室', farm: '☐ 农场',
}
export default function Me() {
  const { state, dispatch } = useUser()
  const navigate = useNavigate()
  const [editing, setEditing] = useState(false)
  const [name, setName] = useState(state.name)
  const myRuns = (state.runHistory || []).map((r) => ({
    name: `${state.name} · ${CO_NAMES[r.cold] || r.coName || '公司'}`,
    score: runScore(r), avatar: state.avatar, me: true, run: r,
  }))
  const rankList = [...RANK_BASE, ...myRuns].sort((a, b) => b.score - a.score)
  const myBest = myRuns.length ? myRuns.reduce((a, b) => (b.score > a.score ? b : a)) : null
  const myRank = myBest ? rankList.findIndex((r) => r.me && r.score === myBest.score) + 1 : null
  const coBest = {}
  ;(state.runHistory || []).forEach((r) => {
    if (!coBest[r.cold] || runScore(r) > runScore(coBest[r.cold])) coBest[r.cold] = r
  })
  const coCompare = Object.values(coBest).sort((a, b) => runScore(b) - runScore(a))
  const earned = BADGES.filter((b) => state.badges.includes(b.id))
  const locked = BADGES.filter((b) => !state.badges.includes(b.id))
  const saveName = () => {
    dispatch({ type: 'SET_NAME', name: name || '小会计' })
    setEditing(false)
  }
  return (
    <div className="page fade-in">
      <div className="card" style={{ textAlign: 'center' }}>
        <div style={{ fontSize: 48 }}>{state.avatar}</div>
        {editing ? (
          <div className="mt12">
            <input
              value={name}
              onChange={(e) => setName(e.target.value)}

```

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```

      style={{ padding: 10, borderRadius: 10, border: '2px solid var(--line)', textAlign: 'center', fontWeight: 700
    />
    <button className="btn mt12" onClick={saveName}>保存</button>
  </div>
) : (
  <div>
    <div style={{ fontWeight: 800, fontSize: 18 }}>{state.name}</div>
    <div style={{ color: 'var(--text-soft)', fontSize: 13 }}>Lv. {state.level} · {TIERS.find((t) => t.id === state.t

```

```

        <button className="btn ghost mt12" style={{ width: 'auto', padding: '6px 14px', fontSize: 13 }} onClick={() => s
    </div>
  })
  <div className="flex gap8 mt20">
    <div className="pill coins" style={{ flex: 1, justifyContent: 'center' }}>□ {state.coins}</div>
    <div className="pill level" style={{ flex: 1, justifyContent: 'center' }}>□ 经验 {state.exp}</div>
  </div>
</div>
<div className="section-title">□ 我的职称</div>
<div className="flex gap8" style={{ flexWrap: 'wrap' }}>
  {TIERS.map((t) => (
    <button
      key={t.id}
      className={`btn ${state.tier === t.id ? '' : 'ghost'}`}
      style={{ width: 'auto', padding: '8px 14px', fontSize: 13 }}
      onClick={() => dispatch({ type: 'SET_TIER', tier: t.id })}
    >
      {t.name}
    </button>
  ))}
</div>
<div className="section-title mt20">□ 徽章 ({earned.length}/{BADGES.length})</div>
<div className="badge-grid">
  {earned.map((b) => (
    <div key={b.id} className="badge-item" title={b.desc}>
      <div className="b-emoji">{b.emoji}</div>
      <div className="b-name">{b.name}</div>
    </div>
  ))}
  {locked.map((b) => (
    <div key={b.id} className="badge-item locked" title={b.desc}>
      <div className="b-emoji">□</div>
      <div className="b-name">{b.name}</div>
    </div>
  ))}
</div>
<div className="section-title mt20">□ 经营排行榜（本地）</div>
<div className="card" style={{ padding: 8 }}>
  {rankList.map((r, i) => (
    <div key={i} className={`rank-row` + (r.me ? ' me' : '')}>
      <span className="rk">{i + 1}</span>
      <span style={{ fontSize: 22 }}>{r.avatar}</span>

```

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```

    <span className="nm">{r.name} {r.me ? '（我）' : ''}</span>
    <span className="sc">{r.score}</span>
  </div>

```

```

    ))}
    <div style={{ textAlign: 'center', color: 'var(--text-soft)', fontSize: 12, marginTop: 6 }}>
      {myRank ? `我的排名：第 ${myRank} 名 · 加油冲榜! 🏆` : `多玩几局就能登上排行榜啦~`}
    </div>
  </div>
  /* G: 多公司横向对比 —— 各类公司已体验的最佳一局 */
  <div className="section-title mt20">🏢 多公司对比</div>
  {coCompare.length === 0 ? (
    <div className="card" style={{ textAlign: 'center', color: 'var(--text-soft)', fontSize: 13 }}>
      还没有经营记录，去开一家公司吧！
    </div>
  ) : (
    <div className="card" style={{ padding: 8 }}>
      <div style={{ display: 'grid', gridTemplateColumns: '1.4fr 1fr 0.8fr 0.8fr', fontSize: 11, color: 'var(--text-soft)' }}>
        <span>公司类型</span><span>净利润(万)</span><span>星级</span><span>月份</span>
      </div>
      {coCompare.map((r, i) => (
        <div key={i} style={{ display: 'grid', gridTemplateColumns: '1.4fr 1fr 0.8fr 0.8fr', alignItems: 'center', padding: 8 }}>
          <span>{CO_NAMES[r.colId] || r.coName}</span>
          <span style={{ fontWeight: 700, color: (r.profit || 0) >= 0 ? 'var(--primary-deep)' : '#E5484D' }}>{r.profit}</span>
          <span>{'🌟'.repeat(r.stars || 0)}</span>
          <span>{r.months} 月</span>
        </div>
      ))}
      <div style={{ fontSize: 11, color: 'var(--text-soft)', marginTop: 6, lineHeight: 1.6 }}>
        提示：用不同公司类型、不同纳税人身份或出资方式多开几局，对比哪种组合最赚钱——这就是“平行经营”的乐趣。
      </div>
    </div>
  )}
  <button className="btn ghost mt12" onClick={() => navigate('/')}>🏠 返回首页</button>
  <button className="btn ghost mt12" onClick={() => { if (confirm('确定重置所有进度?')) dispatch({ type: 'RESET' }) }}>
    🔄 重置进度
  </button>
</div>
)
}
/* ===== 文件: pages/Reports.jsx ===== */
import { useState, useRef } from 'react'
import { COMPANIES } from '../data/companies.js'
import { createCompany, applyBusiness, monthEnd, buildReports, applyFunding, applyTaxType, vatOnSale, vatOnPurchase, settle } from '../utils'
import ReportView from '../components/ReportView.jsx'
export default function Reports() {
  const [colId, setColId] = useState('shop')
  const [diffId, setDiffId] = useState('easy')
  const [report, setReport] = useState(null)
  const [reportReady, setReportReady] = useState(false)
  const reportRef = useRef(null)

```

```

const generate = () => {
  const co = COMPANIES.find((c) => c.id === cold)
  const s = createCompany(cold, diffId)
  applyFunding(s, 'full')
  applyTaxType(s, 'small')
  co.fixedAssets.forEach((fa) => applyBusiness(s, [
    { side: 'debit', account: '固定资产', amount: fa.value },
    { side: 'credit', account: '银行存款', amount: fa.value },
  ], `购置${fa.name}`))
  for (let m = 1; m <= 2; m++) {
    const ds = co.economics.dealSize * s.scale
    applyBusiness(s, vatOnPurchase(s, ds, false).entries, '进货')
    applyBusiness(s, vatOnSale(s, ds, co.revenueAccount).entries, '卖货')
    applyBusiness(s, co.costOfSale(ds * (1 - co.economics.margin)), '结转成本')
    applyBusiness(s, [{ side: 'debit', account: '管理费用-房租', amount: co.economics.rent * s.scale }, { side: 'credit',
    applyBusiness(s, [{ side: 'debit', account: '管理费用-工资', amount: co.economics.salary * s.scale }, { side: 'credit',
    monthEnd(s)
  }
  settleTax(s)
  setReport(buildReports(s))
  setReportReady(true)
}
return (
  <div className="page fade-in">
    <div className="card">
      <div className="section-title" style={{ marginTop: 0 }}>□ 财务报表中心</div>
      <div className="sub-title">选公司类型，一键生成真实三张表（含平衡校验）</div>
      <div className="flex gap8" style={{ flexWrap: 'wrap' }}>
        {COMPANIES.map((c) => (
          <button key={c.id} className={`btn ghost ${cold === c.id ? '' : 'ghost'}`} style={{ width: 'auto', padding: '8px
        ))}
      </div>
      <div className="flex gap8 mt12">
        {Object.values(DIFFICULTY).map((d) => (
          <button key={d.id} className="btn ghost" style={{ width: 'auto', padding: '8px 12px' }} onClick={() => setDiffI
        ))}
      </div>
      <button className="btn mt12" onClick={() => { generate(); setTimeout(() => reportRef.current?.scrollIntoView({ behav
    </div>
    {report ? (
      <div ref={reportRef}>
        <ReportView report={report.balance} />
        <ReportView report={report.income} />
      </div>
    ) : (
      <div className="card" style={{ textAlign: 'center', color: 'var(--text-soft)' }}>
        点击「生成报表」查看资产负债表与利润表
      </div>
    )
  )

```

```
    })  
  </div>  
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```

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```
    )  
  }  
/* ===== 文件: store/UserContext.jsx ===== */  
import React, { createContext, useContext, useEffect, useReducer } from 'react'  
const STORAGE_KEY = 'accounting_game_user_v1'  
const defaultState = {  
  name: '小会计',  
  avatar: '□',  
  coins: 0,  
  exp: 0,  
  level: 1,  
  tier: 'primary', // primary | middle | senior  
  completedLevels: [], // 历史保留字段（兼容）  
  badges: [], // badge ids earned  
  gameBestProfit: 0,  
  companiesRun: [], // 已体验的公司类型 id  
  runHistory: [], // H: 每局通关记录 { cold, coName, profit, stars, months, date }  
}  
function reducer(state, action) {  
  switch (action.type) {  
    case 'ADD_COINS':  
      return { ...state, coins: state.coins + action.amount }  
    case 'ADD_EXP': {  
      const exp = state.exp + action.amount  
      const level = Math.floor(exp / 100) + 1  
      return { ...state, exp, level }  
    }  
    case 'COMPLETE_LEVEL': {  
      const done = state.completedLevels.includes(action.id)  
      ? state.completedLevels  
      : [...state.completedLevels, action.id]  
      return { ...state, completedLevels: done }  
    }  
    case 'EARN_BADGE': {  
      if (state.badges.includes(action.id)) return state  
      return { ...state, badges: [...state.badges, action.id] }  
    }  
    case 'SET_GAME_BEST': {  
      const best = Math.max(state.gameBestProfit, action.profit)  
      let badges = state.badges  
      if (best >= 500 && !badges.includes('boss500')) badges = [...badges, 'boss500']  
      if (best >= 2000 && !badges.includes('tycoon')) badges = [...badges, 'tycoon']  
      return { ...state, gameBestProfit: best, badges }  
    }  
  }  
}
```

```

    }
    case 'SET_TIER':
      return { ...state, tier: action.tier }
    case 'COMPANY_RUN': {
      if (state.companiesRun.includes(action.id)) return state
      return { ...state, companiesRun: [...state.companiesRun, action.id] }
    }
  }
}
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```

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```

    case 'RECORD_RUN': {
      const run = {
        cold: action.cold,
        coName: action.coName,
        profit: +(action.profit || 0).toFixed(2),
        stars: action.stars || 0,
        months: action.months || 0,
        date: action.date || new Date().toISOString().slice(0, 10),
      }
      return { ...state, runHistory: [...state.runHistory, run] }
    }
    case 'SET_NAME':
      return { ...state, name: action.name, avatar: action.avatar || state.avatar }
    case 'RESET':
      return { ...defaultState }
    default:
      return state
  }
}
}
const UserContext = createContext(null)
export function UserProvider({ children }) {
  const [state, dispatch] = useReducer(reducer, defaultState, (init) => {
    try {
      const raw = localStorage.getItem(STORAGE_KEY)
      if (raw) return { ...defaultState, ...JSON.parse(raw) }
    } catch (e) {}
    return init
  })
  useEffect(() => {
    try { localStorage.setItem(STORAGE_KEY, JSON.stringify(state)) } catch (e) {}
  }, [state])
  return (
    <UserContext.Provider value={{ state, dispatch }}>{children}</UserContext.Provider>
  )
}
export function useUser() {
  const ctx = useContext(UserContext)
  if (!ctx) throw new Error('useUser must be used within UserProvider')
}

```

```

    return ctx
}
/* ===== 文件: styles/global.css ===== */
:root {
  --bg-soft: #FFF7EC;
  --bg-card: #FFFFFF;
  --primary: #FF8A5B;
  --primary-deep: #F2693F;
  --accent: #4ECDC4;
  --accent-deep: #2BB3A8;
  --gold: #FFC93C;
  --text: #3D3A4B;
}
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```

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```

  --text-soft: #8B89A0;
  --line: #F0E6D8;
  --debit: #FF8A5B;
  --credit: #4ECDC4;
  --shadow: 0 8px 24px rgba(255, 138, 91, 0.12);
  --shadow-strong: 0 12px 32px rgba(255, 138, 91, 0.22);
  --radius: 20px;
  --radius-sm: 12px;
}
html, body, #root { height: 100%; }
body {
  font-family: "PingFang SC", "Microsoft YaHei", "Segoe UI", system-ui, sans-serif;
  background: linear-gradient(160deg, #FFF7EC 0%, #FFEFE0 100%);
  color: var(--text);
  -webkit-font-smoothing: antialiased;
}
button {
  font-family: inherit;
  cursor: pointer;
  border: none;
  outline: none;
}
.app-shell {
  min-height: 100%;
  display: flex;
  flex-direction: column;
  max-width: 480px;
  margin: 0 auto;
  position: relative;
  background: linear-gradient(160deg, #FFF7EC 0%, #FFEFE0 100%);
}
.topbar {
  position: sticky;
}

```

```

    top: 0;
    z-index: 20;
    display: flex;
    align-items: center;
    gap: 12px;
    padding: 14px 18px;
    background: rgba(255, 247, 236, 0.92);
    backdrop-filter: blur(8px);
}
.topbar .title { font-size: 18px; font-weight: 800; letter-spacing: 1px; }
.topbar .spacer { flex: 1; }
.pill {
    display: inline-flex;
    align-items: center;
    gap: 6px;
    background: var(--bg-card);
    border-radius: 999px;
}
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```

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```

    padding: 6px 12px;
    font-weight: 700;
    font-size: 13px;
    box-shadow: var(--shadow);
}
.pill.coins { color: var(--primary-deep); }
.pill.level { color: var(--accent-deep); }
.page { flex: 1; padding: 18px 18px 96px; }
.section-title {
    font-size: 20px;
    font-weight: 800;
    margin: 6px 0 14px;
}
.sub-title { color: var(--text-soft); font-size: 13px; margin-bottom: 16px; }
.card {
    background: var(--bg-card);
    border-radius: var(--radius);
    box-shadow: var(--shadow);
    padding: 18px;
    margin-bottom: 14px;
}
.btn {
    display: inline-flex;
    align-items: center;
    justify-content: center;
    gap: 8px;
    background: var(--primary);
    color: #fff;
}

```

```

    font-weight: 800;
    font-size: 15px;
    padding: 13px 18px;
    border-radius: 16px;
    box-shadow: var(--shadow);
    transition: transform .12s ease, box-shadow .12s ease;
    width: 100%;
}
.btn:active { transform: scale(0.96); }
.btn.secondary { background: var(--accent); }
.btn.ghost { background: #fff; color: var(--primary-deep); box-shadow: inset 0 0 0 2px var(--line); }
.btn.disabled { opacity: .5; cursor: not-allowed; }
.bottom-nav {
    position: fixed;
    bottom: 0;
    left: 50%;
    transform: translateX(-50%);
    width: 100%;
    max-width: 480px;
    display: flex;
    background: var(--bg-card);
    box-shadow: 0 -6px 20px rgba(0,0,0,0.06);
}
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```

```

    border-radius: 24px 24px 0 0;
    padding: 8px 0 calc(8px + env(safe-area-inset-bottom, 0px));
    z-index: 30;
}
.bottom-nav .nav-item {
    flex: 1;
    display: flex;
    flex-direction: column;
    align-items: center;
    gap: 3px;
    font-size: 11px;
    color: var(--text-soft);
    background: none;
    font-weight: 600;
}
.bottom-nav .nav-item.active { color: var(--primary-deep); }
.bottom-nav .nav-item.emoji { font-size: 22px; }
.progress-track {
    height: 10px;
    background: var(--line);
    border-radius: 999px;
    overflow: hidden;
}

```

```
.progress-fill {
  height: 100%;
  background: linear-gradient(90deg, var(--gold), var(--primary));
  border-radius: 999px;
  transition: width .5s ease;
}
.level-card {
  position: relative;
  overflow: hidden;
}
.level-card .lv-head {
  display: flex;
  align-items: center;
  gap: 12px;
}
.level-card .lv-icon {
  width: 46px; height: 46px;
  border-radius: 14px;
  display: grid; place-items: center;
  font-size: 24px;
  background: var(--bg-soft);
  flex-shrink: 0;
}
.level-card .lv-name { font-weight: 800; font-size: 16px; }
.level-card .lv-desc { color: var(--text-soft); font-size: 12px; margin-top: 2px; }
.level-card .lv-badge {
  margin-left: auto;
}
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```

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```
font-size: 11px;
font-weight: 700;
padding: 4px 10px;
border-radius: 999px;
}
.lv-badge.done { background: #E4F8F1; color: var(--accent-deep); }
.lv-badge.lock { background: #F2EEE9; color: var(--text-soft); }
.lv-badge.todo { background: #FFEFE2; color: var(--primary-deep); }
.chip {
  display: inline-block;
  padding: 5px 12px;
  border-radius: 999px;
  font-size: 12px;
  font-weight: 700;
  background: var(--bg-soft);
  color: var(--primary-deep);
  margin: 0 6px 6px 0;
}
```

```
.tier-bar {
  display: flex;
  align-items: center;
  gap: 8px;
  margin: 18px 0 10px;
  font-weight: 800;
  font-size: 15px;
}
.tier-bar .dot { width: 10px; height: 10px; border-radius: 50%; }
.tier-bar.primary .dot { background: var(--accent); }
.tier-bar.middle .dot { background: var(--primary); }
.tier-bar.senior .dot { background: var(--gold); }
.option {
  display: flex;
  align-items: center;
  gap: 12px;
  padding: 14px 16px;
  border-radius: var(--radius-sm);
  background: var(--bg-soft);
  margin-bottom: 10px;
  font-weight: 600;
  border: 2px solid transparent;
  transition: all .15s ease;
}
.option.selected { border-color: var(--primary); background: #FFF1E8; }
.option.correct { border-color: var(--accent); background: #E4F8F1; }
.option.wrong { border-color: #FF6B6B; background: #FFEDED; }
.option .opt-key {
  width: 26px; height: 26px;
  border-radius: 50%;
  background: #fff;
  display: grid; place-items: center;
}
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```

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```
font-weight: 800;
font-size: 13px;
flex-shrink: 0;
}
.option.correct .opt-key { background: var(--accent); color: #fff; }
.option.wrong .opt-key { background: #FF6B6B; color: #fff; }
.entry-row {
  display: flex;
  align-items: stretch;
  gap: 10px;
  margin-bottom: 10px;
}
.entry-side {
```

```
flex: 1;
border-radius: var(--radius-sm);
padding: 12px 14px;
position: relative;
opacity: 0;
transform: translateY(14px);
animation: pop-in .5s ease forwards;
}
.entry-side.debit { background: #FFF1E8; border-left: 5px solid var(--debit); }
.entry-side.credit { background: #E9F8F6; border-left: 5px solid var(--credit); }
.entry-side.label { font-size: 12px; font-weight: 800; letter-spacing: 1px; }
.entry-side.debit.label { color: var(--primary-deep); }
.entry-side.credit.label { color: var(--accent-deep); }
.entry-side.acc { font-size: 15px; font-weight: 700; margin-top: 4px; }
.entry-side.amt { margin-left: auto; font-weight: 800; }
.entry-amount-line {
  display: flex; align-items: center; justify-content: center;
  font-weight: 800; color: var(--text-soft);
}
@keyframes pop-in {
  to { opacity: 1; transform: translateY(0); }
}
.delay-1 { animation-delay: .15s; }
.delay-2 { animation-delay: .3s; }
.delay-3 { animation-delay: .45s; }
.toast {
  position: fixed;
  left: 50%;
  bottom: 110px;
  transform: translateX(-50%);
  background: var(--text);
  color: #fff;
  padding: 10px 18px;
  border-radius: 999px;
  font-weight: 700;
  font-size: 14px;
  z-index: 50;
}
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```

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```
box-shadow: var(--shadow-strong);
animation: toast-in .3s ease;
}
@keyframes toast-in { from { opacity: 0; transform: translate(-50%, 10px); } }
.shop-stat {
  display: flex;
  justify-content: space-between;
  background: var(--bg-soft);
}
```

```
padding: 10px 14px;
border-radius: var(--radius-sm);
margin-bottom: 8px;
font-weight: 700;
}
.shop-stat.v { color: var(--primary-deep); }
.modal-mask {
position: fixed; inset: 0;
background: rgba(61, 58, 75, 0.45);
display: grid; place-items: center;
z-index: 60; padding: 20px;
}
.modal {
background: #fff;
border-radius: var(--radius);
padding: 22px;
width: 100%;
max-width: 360px;
box-shadow: var(--shadow-strong);
animation: pop-in .3s ease;
}
.badge-grid { display: grid; grid-template-columns: repeat(3, 1fr); gap: 12px; }
.badge-item {
text-align: center;
background: var(--bg-soft);
border-radius: var(--radius-sm);
padding: 14px 6px;
}
.badge-item.locked { filter: grayscale(1); opacity: .45; }
.badge-item.b-emoji { font-size: 30px; }
.badge-item.b-name { font-size: 12px; font-weight: 700; margin-top: 4px; }
.rank-row {
display: flex; align-items: center; gap: 12px;
padding: 12px 14px;
border-radius: var(--radius-sm);
background: var(--bg-soft);
margin-bottom: 8px;
}
.rank-row.me { background: #FFF1E8; border: 2px solid var(--primary); }
.rank-row.rk { width: 26px; text-align: center; font-weight: 800; }
.rank-row.nm { flex: 1; font-weight: 700; }
.rank-row.sc { color: var(--primary-deep); font-weight: 800; }
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```

```
.hero {
background: linear-gradient(135deg, var(--primary), var(--gold));
border-radius: var(--radius);
```

```
padding: 22px;
color: #fff;
box-shadow: var(--shadow-strong);
margin-bottom: 16px;
}
.hero h1 { font-size: 24px; font-weight: 900; margin-bottom: 6px; }
.hero p { font-size: 13px; opacity: .92; }
.flex { display: flex; }
.between { justify-content: space-between; }
.center { align-items: center; }
.gap8 { gap: 8px; }
.mt12 { margin-top: 12px; }
.mt20 { margin-top: 20px; }
.fade-in { animation: toast-in .4s ease; }
```

